

Board Ease: Boarding House Management and Tenant App

A capstone Project

Presented to the Faculty of Computer Science and Information Technology

NEGROS ORIENTAL STATE UNIVERSITY

Dumaguete City, Philippines

In Partial Fulfillment

Of the Requirements for the Degree of

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

By

Onio, Mercedita

Patajo, Rex Clint

Renquijo, Joemar

Yguinto, Kate Danielle

January 2025

CHAPTER I

INTRODUCTION



Project Context

Technology has significantly reshaped industries across the globe, with the housing rental industry being no exception. With digital solutions, the process of finding and managing rental homes has become faster, efficient, and accessible. Gunawan (2021) emphasized the importance of a digital system that can simplify housing searches, benefiting both tenants and property owners. Innovations in Technology, especially through e-commerce platforms, have revolutionized the housing rental industry by automating essential processes such as marketing, leasing, and sales. These innovations are crucial in modernizing the sector, ensuring its sustainability, and allowing it to thrive in today's technology-driven global market (Ombati, 2022). With the rise of online platforms, tenants can now easily find boarding house listings that match their preferences, while property owners can manage their boarding house listings and reach a wider audience. This digital transformation has not only improved convenience but also enhanced transparency, allowing users to access real-time information and make informed decisions. As the demand for smart solutions in housing grows, the integration of technology will continue to redefine the way the rental industry operates.

The rental housing sector has seen significant growth despite the challenges associated with its operations, which remain inefficient for many. Conventional methods, such as manually searching for advertisements on social media platforms, relying on posters, or using word-of-mouth to inquire about vacancies, are still widely used. These approaches can be time-consuming, frustrating, and often limited to specific locations, leading to missed opportunities for tenants to

find the most suitable accommodations. For boarding house owners, managing vacancies, responding to inquiries from potential tenants, and maintaining records are ongoing challenges. Without an automated system, issues such as miscommunication, double bookings, maintenance delays, and overlooked tenant concerns frequently arise. These problems can negatively affect tenant satisfaction, harm the business's reputation, and hinder long-term success. A more centralized and technology-driven approach is necessary to address these inefficiencies effectively.

This study proposes the development of a boarding house management system called "StayNest." The goal of StayNest is to develop a user-friendly platform that simplifies the process of finding and renting boarding houses, specifically in Dumaguete City, Negros Oriental. Prospective tenants can easily search for available boarding houses using filters such as location, price range, amenities, and other preferences. This simplified search functionality enhances the convenience and efficiency of the house-hunting experience. Additionally, StayNest gives boarding house owners the tools they need to efficiently run their establishments. In order to improve operational efficiency and optimize administrative work, key features include the ability to post, modify, and update listings, contact with potential renters, reply to queries, track payments, and manage bookings.

Purpose and Description

The StayNest Boarding House Management System is a user-friendly platform designed to automate the process of finding and renting boarding houses. By replacing traditional, inefficient methods with a modern, automated solution, StayNest addresses the challenges faced by both tenants and owners. Key features, such as search and filter functionality, an integrated map, listing management tools, booking and inquiry management, a real-time chat feature, payment tracking,

and receipt generation, enhance convenience and operational efficiency for all users. This integrated platform creates a smooth and efficient experience for both tenants seeking suitable accommodations and boarding house owners managing their properties.

Objectives of The Study

The study focuses on achieving the following general and specific objectives to implement the StayNest Boarding House Management System.

General Objectives

The general objective of StayNest is to develop a comprehensive, user-friendly system that simplifies and modernizes the process of finding and renting boarding houses. By integrating advanced features and automation, the system aims to enhance convenience, improve operational efficiency, and bridge the gap between tenants and boarding house owners.

Specific Objectives

This study aims to achieve the following specific objectives:

1. To analyze and gather the valuable requirements and challenges of tenants, boarding house owners, and administrator to translate these into functional features for the development of the proposed system.
2. To design a system incorporating essential features, such as search functionality, booking notifications, messaging tools, and administration panel.
3. To develop a fully functional system that meets the identified requirements and incorporates key features.

4. To implement and deploy the system, ensuring its functionality, security, and user-friendliness for all intended users.
5. To test the Boarding House Management System in Dumaguete City, ensuring functionality, reliability, and user satisfaction.
6. To perform regular system maintenance such as monitoring, issue resolution, performance enhancements, and implementing improvements based on user feedback.

Scope and Limitations

This section defines the scope and limitations of the StayNest Boarding House Management System project, detailing the features to be included and the potential constraints and challenges that may impact its implementation and effectiveness.

Scope

The scope of the StayNest: Boarding House Management System defines its features and functionalities, catering to its primary users: prospective tenants, tenants, boarding house owners, and administrator. Prospective tenants can use the system to search and explore boarding house options within Dumaguete City, utilizing an integrated map for geographic guidance and search filters to help tenants find a boarding house that matches their preferences, while offline inspections and tenant-owner negotiations remain part of the process. Tenants benefit from communication tools, regular updates about their chosen boarding houses, and the convenience of an online payment system for rent transactions. Property owners can efficiently manage listings and property details, interact with prospective and current tenants, track payments, and generate

receipts. For administrators, the system provides tools to monitor user activities, ensure proper platform functionality, and maintain smooth operations.

Limitations

The StayNest Boarding House Management System has several limitations. It is specifically designed for tenants seeking boarding houses in Dumaguete City and cannot accommodate users searching in other locations. Tenants still need to visit properties to inspect them and negotiate with owners in person. The accuracy of information, such as room availability and pricing, depends on how often property owners update their boarding house listings. Online payment options are limited to GCash, which may not suit all users. Owners might provide incorrect or misleading details about their properties, leading to potential tenant dissatisfaction. Collecting user and owner data involves risks of misuse or breaches, making strong security measures essential. Additionally, the system's search filters may not fully address all tenant preferences, such as budget, location, or specific amenities, making it harder to find suitable accommodations.

Significance of The Project

The StayNest Boarding House Management System proposed in this study offers significant benefits to various stakeholders involved in the management and operation of boarding houses. The significance of this project is evident in its potential positive impact on the following:

Boarding House Owners. StayNest enables boarding house owners to efficiently manage listings, track bookings, and communicate with tenants. By automating tasks like booking

confirmations, inquiries, and payments, the system reduces administrative workload, minimizes errors, and improves operational efficiency, leading to enhanced profitability and business growth.

Property Management Staff. StayNest enhances efficiency and reduces errors in managing boarding house operations. It allows them to focus more on delivering excellent service to both tenants and owners.

Tenants. StayNest enhances the boarding house experience by simplifying the process of finding available accommodations. The system allows easy browsing of boarding house listings, booking of rooms, and communication with owners, improving tenant satisfaction.

Prospective Tenants. StayNest provides prospective tenants with an efficient platform to search for available boarding houses based on specific preferences, such as location, price, and amenities. The system's easy-to-use interface, combined with detailed boarding house listings and real-time availability, helps tenants make well-informed decisions quickly and efficiently, reducing the time and effort required to find suitable accommodations.

Future Researchers. This project serves as a valuable case study in the development of management systems for the real estate and housing industries. The technical insights and lessons learned from its development can contribute to further studies in IT solutions for property management and inspire similar projects in related fields.

CHAPTER II



REVIEW OF RELATED LITERATURE AND STUDIES

This chapter provides a complete assessment of the related literature and studies relevant to the current study. It aims to investigate and analyze prior findings and advancements, using such information to establish a solid basis for the current study.

Review of Related Literature

Finding suitable housing is a significant challenge for individuals living away from home, especially students and workers relocating to new areas. According to Effendy et al. (2020), a safe, comfortable, and conveniently located living space can enhance academic and professional success, while inadequate housing can cause stress and physical or mental health issues. Despite its importance, the process of finding appropriate boarding houses remains challenging due to limited information, time constraints, and mobility issues. Factors such as facilities, price, location, security, reputation, services, and advertising significantly influence the decision-making process for tenants.

In response to these challenges, Adams (2020) emphasized that choosing a suitable boarding house is one of the most critical decisions tenants face, as the comfort and convenience of their environment greatly affect their well-being and productivity. However, the search for ideal accommodations is often complicated by individual preferences and the availability of options that meet specific needs. Maalsen and Gurran (2022) further elaborated on these complexities, noting that factors such as rental prices, proximity to workplaces or schools, amenities, and living conditions play a crucial role in tenant satisfaction.

The limitations of traditional search methods further heighten these challenges. Gunawan (2021) pointed out that students and workers often face difficulties in finding suitable boarding houses due to inefficiencies in traditional methods and a lack of accessible information on critical aspects such as location, rent, facilities, and property owners. These limitations frequently force tenants to rely on word-of-mouth recommendations or time-consuming manual searches. Gunawan highlighted the need for a centralized digital solution to streamline the process, making housing searches faster and more efficient for both tenants and property owners.

Building on this need, **several studies propose digital solutions** to centralize information and improve accessibility. Rachmadi et al. (2021) discussed the challenges faced by students in Yogyakarta, particularly those unfamiliar with the area and lacking personal contacts. The study suggested an online boarding house information system that allows property owners to advertise their vacancies, helping tenants access detailed and accurate information. Both Gunawan and Rachmadi underscore the importance of technology in making the search process more efficient for tenants while empowering property owners to manage their listings effectively.

Technological innovations, particularly e-commerce platforms, are transforming the rental housing industry by automating processes such as marketing, leasing, and sales. These digital solutions improve accessibility, streamline transactions, and centralize property information while offering features like real-time updates, virtual tours, and interactive interfaces. By enhancing user experience and decision-making, these platforms democratize access to real estate, expand market reach, and increase customer engagement. Such innovations play a critical role in modernizing the industry and ensuring its sustainability in a technology-driven global market (Ombati, 2022).

Anwar and Purnama (2022) demonstrated the use of structured database systems based on the Database Life Cycle (DBLC) approach, facilitating robust information system creation. These systems enabled property owners to promote their listings effectively while providing tenants with comprehensive details about available accommodations. Similarly, Wahyudi (2023) demonstrated the effectiveness of web-based boarding house systems in addressing common challenges in the rental service industry. Using a structured development process, the system provided an intuitive platform for tenants to search for accommodations and for property owners to market their listings. By reducing the time and effort required to find suitable housing, this approach benefits all stakeholders.

Boarding house owners often face challenges in effectively advertising their properties, while tenants struggle to find suitable accommodations using traditional methods like posters or tarpaulins, which have limited reach. To address these issues, a two-platform system was proposed, comprising a mobile app for Android users and a web-based platform for property owners. The mobile app allowed tenants in Purwokerto to browse detailed information about boarding houses, including prices, amenities, and availability, while the web platform enabled owners to easily update and manage their property listings. This system simplified the search process for tenants, broadened the reach for owners, and minimized effort, showcasing the role of technology in enhancing accessibility, improving user experience, and modernizing the traditional boarding house industry (Guntur et al., 2024).

The integration of advanced technologies such as the Internet of Things (IoT) and artificial intelligence (AI) has further transformed boarding house management. Gormantara et al. (2024) explored the use of IoT to modernize traditional systems, enabling real-time data exchange, automation of tasks, and improved operational efficiency. Features such as utility monitoring,

room availability checks, and tenant communication were automated, enhancing service quality and reducing manual effort.

IoT technologies play a transformative role in improving efficiency and effectiveness in various sectors by enabling real-time data exchange and automated control. In the housing sector, IoT facilitates the development of smart, user-friendly systems that integrate real-time information, remote access, and automation to address challenges in information management and user engagement. These adaptive, technology-driven solutions are designed to meet user needs while streamlining processes and enhancing resource management (Rubaeah et al., 2023). Additionally, by integrating automation with advanced technologies such as AI, IoT, and ML, organizations improve performance, enhance service precision, and secure a competitive advantage in the global economy (Luo, 2023).

Database-driven systems have been instrumental in enhancing booking and payment processes. Gonzalez et al. (2020) designed a reservation system that uses database technology to manage room bookings and tenant data. By providing accurate and fast tracking of occupancy and transactions, the system improves customer satisfaction while reducing administrative workload. Wang et al. (2022) introduced a mobile application that focuses on simplifying operations like room reservations, maintenance requests, and tenant communications, further highlighting the efficiency of mobile solutions in modern boarding house management. From mobile and web applications to IoT and AI-integrated systems, the advancements in boarding house management technologies showcase a wide range of innovations. These studies collectively underscore the importance of leveraging technology to improve both the tenant experience and property management processes, offering scalable and efficient solutions to meet the evolving demands of the accommodation industry. Boarding houses play a crucial role in providing affordable and

accessible accommodations for students and professionals near schools, universities, or workplaces. As urban areas continue to grow and the demand for convenient housing options increases, the need for innovative solutions to address these challenges has become more pressing. In response, numerous technological advancements have been developed to simplify the process of finding and managing boarding houses. These systems aim to make housing searches more accessible for tenants while also offering tools that streamline operations for property owners. For tenants, these systems provide comprehensive property details, making it easier to make informed decisions. Information such as location, pricing, and amenities is often displayed in user-friendly formats, allowing users to compare options that best suit their needs. Advanced search filters, maps, and reviews further enhance the decision-making process, saving time and effort. For property owners, these systems offer tools to manage listings more efficiently, such as real-time availability updates, rental tracking, and automated payment solutions. By bridging the gap between tenants and landlords, these innovations not only improve the accessibility of housing but also foster greater transparency and efficiency in the housing market (Hamzah et al., 2022).

Traditionally, managing boarding houses involves a lot of manual tasks like listing rooms, answering inquiries, and handling bookings. However, using web-based systems can make these processes faster and more efficient. Studies show that online systems in property management help reduce manual work, allowing owners to manage their properties more easily. Tenants also benefit from the ability to find and book rooms online, making the process simpler and quicker. These systems can also improve customer service, as they help property owners stay organized and update room availability in real time. Overall, adopting these systems not only makes boarding house management more efficient but also improves the experience for both owners and tenants, reflecting a broader shift towards digital tools in the property industry (Kristianto, S., 2022).

Several studies have proposed innovative solutions to simplify the search for boarding houses. A boarding rental information system was created to simplify booking and payment for tenants while helping property owners manage rentals. Using interviews, observations, and literature reviews, the system was developed using the Rapid Application Development (RAD) model. The system improves tenant convenience by making transactions easier and helps property owners manage and promote their properties more efficiently. The study shows that such a system can simplify rental management, improve operations, and expand the reach of boarding house businesses for both tenants and owners (Helling et al., 2020).

The "Dorm Finder" system, developed by Leo et al. (2023), uses Multi-Attribute Utility Theory (MAUT) to make finding student housing easier and more personalized. This theory helps people make decisions by looking at different factors, ranking their importance, and combining them into a score. In the "Dorm Finder" system, factors like location, price, amenities, and distance to campus are given scores based on what the student values most. For instance, if affordability is a priority, cheaper options will appear higher in the results, while those who care more about being close to campus will see nearby listings prioritized. This personalized approach helps students find housing that matches their specific needs and preferences, reducing stress and saving time. The system is also a valuable tool for property owners. It allows them to highlight the most appealing features of their listings, like affordable prices or convenient locations, making it easier to attract tenants. The platform is simple to use, letting students adjust their preferences easily while updating the results instantly. This ensures a smooth experience for both tenants and landlords. By combining a user-friendly interface with MAUT, "Dorm Finder" creates a smarter and more efficient way to search for housing, making the process faster, more transparent, and more effective for everyone involved.

Further expanding on the impact of web-based solutions, Yuniza (2023) focused on the design of a boarding house information system at Bunda Kosan. The research aimed to address common problems tenants face when searching for housing, especially in cities where the demand for temporary accommodations is high. Instead of having to visit multiple properties in person, the web-based system lets users search, compare, and book boarding houses online. This saves time, reduces costs, and makes the process much easier for tenants. The system also helps property owners in many ways. It allows them to promote their listings to a wider audience, increasing the chances of finding tenants quickly. Features like real-time availability updates, online booking, and secure payment options make managing rentals simpler and more efficient. Additionally, property owners can use the system to keep track of transactions and organize their rental data. Yuniza's work highlights how web-based solutions can make the housing process smoother for both tenants and property owners, improving accessibility and convenience in today's fast-paced world.

Technology-driven management systems play a vital role in overcoming key challenges in the housing and rental industry, particularly in enhancing affordability and accessibility. These systems expand the reach of rental services through digital marketplaces, enabling property managers to connect with a broader audience. By incorporating advanced features such as electronic locks and secure payment options, these systems bolster security while improving user convenience. Additionally, smart technologies integrated into these platforms enhance energy efficiency and optimize resource consumption. Through automation, predictive analytics, and user-friendly interfaces, technology-driven systems enhance information management and decision-making processes. They provide real-time access to critical information, fostering transparency and improving communication between property managers and tenants. This

streamlined interaction leads to greater efficiency and satisfaction for all stakeholders. By focusing on operational improvements and sustainability, these systems are poised to create a more equitable and forward-thinking future for the housing and rental sectors (Wijburg & Waldron, 2020).

Review of Related Studies



To optimize management processes for boarding houses, modern Customer Relationship Management (CRM) tools have been applied in the boarding house industry. Garg (2023) highlighted that CRM systems improve efficiency by automating tasks like tenant management, payment tracking, and property listings. This study developed a property rental system using Salesforce, simplifying tasks such as rent collection and reporting. The system uses Salesforce tools like Lightning, Apex, and Visualforce to reduce errors, cut costs, and ensure timely rent payments. It includes a database for managing property, leases, and rent details. The technical setup requires a 2.0 GHz processor, 2GB RAM, Windows 7, and Microsoft Visual Basic 6. Similarly, Habib & Maryam (2023) utilized CRM to develop a management system for YCMI Character Boarding Houses. Built using tools like XAMPP and Laravel, the system automated manual processes like payment tracking and complaint handling, achieving a 79.11% acceptance rate and significantly improving administrative efficiency.

Emerging technologies like the Internet of Things (IoT) and Artificial Intelligence (AI) are improving how boarding houses are managed. Zhao et al. (2021) introduced "Otaku," a smart dormitory management system leveraging IoT technology to enhance student living environments. This decentralized system integrates server storage, mobile app interfaces, and terminal IoT devices for efficient resource allocation, dormitory access, and monitoring. Built on the Arduino

Mega 2560 and ESP8266 Wi-Fi modules, the system offers a cost-effective solution for dormitory management. It utilizes Wi-Fi for communication and supports decentralized device management with real-time status monitoring. The user evaluation reported high satisfaction, demonstrating system robustness, convenience, and potential scalability for broader campus applications. Kumar et al. (2023) further enhanced boarding house systems by adding AI, which helps automate tasks like booking, payment processing, and tenant interactions. AI algorithms predict things like busy booking times or popular room choices, helping property managers make smarter decisions. Additionally, AI-powered chatbots handle common questions and requests, making communication quicker and easier. These technologies make the system more efficient, improve customer satisfaction, and save time by automating routine tasks. Combining IoT and AI helps create smarter, more responsive systems for both tenants and property managers.

The growing demand for efficient accommodation search and simplifying property management has created a need for innovative solutions. Several Android-based applications have been developed to address these challenges. For example, Widiastuti et al. (2021) developed an Android-based boarding house search application for Soegijapranata Catholic University students to simplify finding accommodations. The application integrates features such as GPS, Google Maps API, Firebase real-time database, and Android Studio for seamless user experience and accurate data presentation. The system uses Java as the programming language and NoSQL databases hosted on Firebase for real-time data management. The app was found to be effective by 90% of users. Similarly, Consignado et al. (2017) developed HAYBOL, an application designed to help users find accommodations while providing property owners with a platform to advertise their listings. The app integrates Google Maps API, GPS, and a web module for administrators to verify business permits and manage listings. Developed using Rapid Application Development

(RAD) methodology, the app was created in Java within the Android Studio environment. It requires internet connectivity and a cloud-based storage system to operate. While the app was praised for its usability, further improvements like expanded coverage and user guides were suggested.

In a related study, Basri et al. (2021) developed a mobile application to assist users in finding and booking dormitories in Karawang, Indonesia, by leveraging Location-Based Service (LBS) technology and the Haversine Formula for calculating the nearest dormitories based on the user's location. The application addresses challenges faced by dormitory seekers in locating nearby accommodations and by property owners in promoting their listings. Developed for Android devices using Java in Android Studio and Firebase as the database, the application features a user-friendly interface that enables users to search for dormitories, view details such as addresses, pictures, facilities, and prices, and book accommodations seamlessly. Additionally, dormitory owners can manage their listings, enhancing their reach to potential tenants. The system was built using the Waterfall development methodology, encompassing requirements analysis, design, implementation, and testing, with the Haversine Formula validated for accuracy against Google Maps. The study demonstrates the effectiveness of integrating LBS technology in mobile applications to improve accessibility and convenience in location-based services.

Prototypes have also contributed to enhancing the accommodation search experience. Negara et al. (2021) created a prototype app to make it easier for individuals in Telang to find and book boarding houses based on their preferences, such as location, price, and amenities. The system assists tenants and landlords to communicate more easily and has nine main features with three types of users. The development process involved architectural design using Use Case Diagrams and Business Process Model Notation (BPMN), followed by data and interface design.

The backend was built using PHP with the Laravel framework, while the interface was created using HTML, CSS, and JavaScript. Tests showed the app worked well on most browsers like Chrome, Opera, and Firefox, though there were small issues on Internet Explorer, Firefox, and Safari. Overall, the app successfully made the process of finding and booking boarding houses faster and more convenient.

Kurnia & Wijaya (2024) developed a web-based boarding house information system that provides real-time updates on room availability, amenities, and pricing, making it easier for tenants to find available rooms and obtain accurate information without needing to contact property owners. This real-time functionality helps tenants quickly identify housing options that match their needs, saving time and reducing confusion. Designed using the Waterfall development method, the system incorporates HTML, PHP, JavaScript, and MySQL to ensure efficient data handling and user-friendly interfaces. The platform supports multimedia elements and responsive layouts for enhanced user engagement. The testing phase confirmed successful functionality, efficient navigation, and accurate information presentation. The system simplifies boarding house management and enhances tenant experience, making it a scalable solution for the evolving rental housing market.

Comprehensive management systems are designed to meet the operational needs of property owners, making tasks easier and reducing errors. Angeles et al. (2022) developed a software system that automates tasks like record-keeping, bookings, payments, and tracking maintenance. By automating these processes, the system reduces human errors, such as incorrect data entry or missed requests. It also makes it easier to update and access tenant information quickly, helping property managers respond faster. On the technical side, the system stores data in a secure, centralized database, which ensures all information is up-to-date and accessible in real-

time. This makes operations smoother and enhances tenant satisfaction. Similarly, Abimanyu et al. (2024) used the Spiral Model to create an integrated management system for boarding houses and homestays, combining web and mobile technologies. The Spiral Model focuses on improving the system through feedback and testing in several stages. This approach allows the system to evolve based on user needs. The system was built with PHP for server-side programming and MySQL for database management to handle dynamic data input and verification, ensuring structured and accurate data recording. The platform supports data input for tenant management, billing, and communication which enhances service efficiency for property managers. These systems show how user-friendly designs and strong development processes help create efficient, effective, and adaptable solutions for property management.

Nguyen et al. (2023), developed a cloud-based system to manage boarding houses. The system combines real-time data processing and cloud storage to improve operations. The system uses a microservices architecture, which allows different parts of the system like booking management, payment processing, and tenant communication to work and grow independently. This makes the system more flexible and easier to maintain. Nguyen et al. followed Agile development methods, enabling them to make continuous improvements based on user feedback. For data storage, they used a NoSQL database like MongoDB, which is well-suited for handling the changing and unstructured data involved in managing a boarding house, such as tenant details, rental agreements, and maintenance requests. This flexible data model helps the system quickly adapt to changing needs and manage the growing amount of data generated as the system expands.

Sari et al. (2024) developed a mobile web-based information system to assist students in finding boarding houses while providing property owners with a platform to market their accommodations effectively. The system integrates Google Maps to display accurate locations and

offers detailed information about boarding houses, including prices, facilities, and contact details. By simplifying the search and booking process, the system reduces the effort required by students and enhances the visibility of boarding houses for owners. The development process employed the System Development Life Cycle (SDLC) methodology, ensuring systematic analysis, design, implementation, and testing. The system was built using PHP for backend functionality, along with HTML, CSS, and JavaScript for frontend development. It utilized MySQL as the database, hosted on XAMPP for local deployment and testing. This integration of modern web technologies highlights the system's potential to streamline accommodation searches and promote effective property management.

Sukri et al. (2022) focused on developing a web-based boarding house information system to address the challenges faced by tenants and property owners in Tampan District, Pekanbaru City. The system streamlines the process of finding and managing boarding accommodations by providing a user-friendly platform for prospective tenants to search for boarding houses, view detailed information, and book rooms online. For property owners, the system includes features for promoting accommodations, managing listings, tracking tenant durations, and viewing booking histories. Built using PHP with the Laravel framework, the system leverages MySQL for database management, hosted on XAMPP for local development and deployment. Frontend development utilized HTML, CSS, and JavaScript to ensure a responsive and engaging user interface. Black-box testing validated the functionality and reliability of the system, demonstrating its potential to enhance the efficiency of finding and managing boarding accommodations.

Synthesis

Both the literature and related studies emphasize the significant challenges tenants, particularly students and workers relocating to unfamiliar areas, face in securing suitable housing. Factors such as location, affordability, security, and amenities heavily influence tenant decision-making, with inadequate housing conditions potentially impacting their academic and professional success (Effendy et al., 2020). Traditional housing search methods, including manual searches and word-of-mouth referrals, often result in inefficiencies due to limited information availability and time constraints (Gunawan, 2021). To address these persistent challenges, the integration of digital solutions has emerged as a critical innovation for modernizing property management systems (Gormantara et al., 2020).

The reviewed studies consistently highlight the transformative role of technology-driven platforms in streamlining the property search and management processes. Digital solutions such as dual-platform systems, mobile apps, and web-based portals enable real-time data exchange, comprehensive property information access, and automated processes for both tenants and property owners (Guntur et al., 2024). These advancements simplify tenant searches by providing transparent information about pricing, availability, and amenities, while property owners benefit from efficient tools for updating listings and managing tenant inquiries. Moreover, the use of IoT technologies facilitates automation, real-time monitoring, and improved communication, significantly reducing manual effort and enhancing operational efficiency (Gormantara et al., 2020).

Collectively, these findings underscore the importance of leveraging technology to foster a transparent, user-friendly, and efficient housing ecosystem. By addressing the evolving demands of tenants and streamlining property management, digital innovations not only improve

accessibility and tenant satisfaction but also optimize business operations for property owners, paving the way for growth and modernization in the housing sector (Yuniza, 2023; Gonzalez et al., 2020).

CHAPTER III

TECHNICAL BACKGROUND

This chapter will provide the theoretical and conceptual framework for developing the Boarding House Management System. Theories and concepts relevant to the design and functionality of the system will be examined on this chapter, providing the guidance in its development, and thereby ensuring a strong and user-centered solution.

Theoretical Framework

In understanding the development and adoption of the StayNest Boarding House Management System, this study is grounded in three well-established theories: The Technology Acceptance Model (TAM), The DeLone and McLean Information Systems (IS) Success Model, and Task-Technology Fit (TTF). These frameworks provide a structured approach to analyzing the key factors that influence the acceptance and widespread adoption of the StayNest system by its primary users: tenants, boarding house owners, and administrator.

Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM), developed by Fred Davis in 1989, is a widely used framework that seeks to explain how users come to accept and use new technologies. TAM posits that two main factors, perceived usefulness (PU) and perceived ease of use (PEOU), are fundamental in determining a user's intention to use technology. Perceived usefulness refers to the degree to which a person believes that using the technology will enhance their job performance, while perceived ease of use is the belief that using the technology will be free of effort. These

perceptions are influential because they impact users' attitudes and behavior toward adopting new technologies. Suppose users believe that a system is valuable and easy to use. In that case, they are more likely to accept and consistently use it, leading to higher adoption rates and long-term success of the technology.

The TAM framework will help StayNest Boarding House Management System assess the factors that influence the system's acceptance by its primary users: tenants, boarding house owners, staffs, and administrators. Perceived usefulness, in this case, refers to how tenants view the system's ability to simplify their search for boarding houses, while for property owners, it is about the system's efficiency in managing listings, handling payments, and improving overall operations. Perceived ease of use will be examined from the perspective of how easily users can navigate StayNest's interface, use its search filters, communicate, and complete transactions. By evaluating these two factors, this study will gain valuable insights into how the features of StayNest can be optimized to increase user engagement and satisfaction, which are critical for the system's adoption.

DeLone and McLean IS Success Model

The DeLone and McLean Information Systems (IS) Success Model, developed by William H. DeLone and Ephraim R. McLean in 1992, provides a comprehensive framework for evaluating the success of information systems. The model identifies six key dimensions that contribute to the overall success of a system: system quality, information quality, service quality, use, user satisfaction, and net benefits. It emphasizes that these factors are interrelated and together determine the effectiveness and impact of an information system. Over time, the model has been

widely accepted and adapted for assessing various types of information systems in different contexts, serving as a benchmark for measuring system performance.

In the context of a boarding house management system, the DeLone and McLean IS Success Model is essential for evaluating how well the system meets the needs of both boarding house owners and tenants. System quality focuses on the technical performance and reliability of the software, ensuring that it functions smoothly without errors. Information quality relates to the accuracy and relevance of the data presented to users, while service quality ensures that support and maintenance are readily available. User satisfaction is crucial, as a satisfied user is more likely to continue using the system, and net benefits highlight the overall advantages gained, such as improved efficiency and increased profitability for owners. By applying this model, developers can ensure that the boarding house management system delivers value to its users and supports business growth.

Task-Technology Fit (TTF)

Task-Technology Fit (TTF) is a theory that evaluates how well a technology supports the tasks it is intended to assist, improving user performance and satisfaction. Introduced by Goodhue and Thompson in 1995, TTF highlights the importance of aligning the features of a technology with the specific needs of its users. The framework suggests that a system is most effective when its functionalities match the requirements of the tasks and the skills of those who use it. This ensures that the technology is both practical and efficient, contributing to better outcomes in various applications.

In the case of a boarding house management system, TTF plays a critical role in ensuring that the system meets the needs of both property owners and tenants. For owners, the system should

simplify tasks such as managing listings, responding to tenant inquiries, and handling bookings. For tenants, it should provide a straightforward way to search for and book properties while offering accurate and relevant information. By designing the system to align with these needs, TTF helps improve user satisfaction, simplifying operations, and encourage greater adoption of the technology, ultimately contributing to the success of the management system.

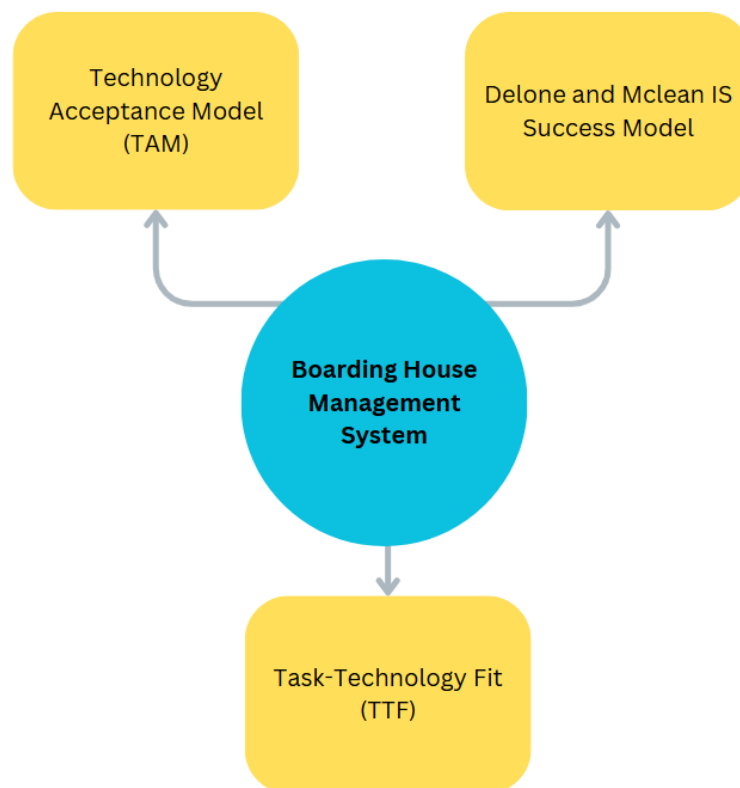


Figure 1  The Theoretical Framework of the Boarding House Management System

Conceptual Framework

Figure 2 illustrates the framework for the StayNest Boarding House Management System, highlighting the key components of Input, Process, and Output. The Input phase gathers essential information, identifying challenges faced by both tenants and boarding house owners, while

reviewing existing solutions and technological trends. This informs the project's requirements. The Process phase uses the Waterfall methodology, which is sequential, ensuring each phase—planning to maintenance—is completed systematically. Finally, the Output phase presents the fully developed StayNest system, designed to meet the needs of tenants and owners, providing a user-friendly platform for efficient property management and streamlined accommodation searches.

Input

The Input phase for developing StayNest involves several crucial steps. First, a thorough understanding of the challenges tenants and boarding house owners face is obtained. This includes identifying the gaps in their current processes such as property listings, booking management, and payment tracking. In addition, the project team conducts interviews with property owners and tenants to gather their input on system requirements. The feedback from potential users ensures that the system will be tailored to meet their needs and preferences. The team also examines best practices in real estate management and current technological solutions to ensure that StayNest incorporates innovative features that align with user expectations. This comprehensive research phase provides the foundation for moving on to the Process phase, where the Waterfall methodology guides the development stages.

Process

In developing StayNest, the project follows the Waterfall methodology, a structured approach that consists of sequential phases. The first phase, Planning, involves defining the project's objectives, identifying key requirements, and allocating resources. The Design phase follows, where the system architecture, user interface, and navigation flow are planned to ensure

ease of use and accessibility for tenants and property owners. During the Development phase, the system is built according to the design specifications, incorporating necessary functionalities like booking systems, messaging tools, and payment tracking. The implementation phase is where the system is deployed for tenants and boarding house owners. This phase includes user training and gathering any feedback for improvements. Once implementation is complete, the system is tested to ensure that the system is functioning correctly and meets the specified requirements, including usability testing with real users to gather feedback. Finally, the Maintenance phase ensures the system is regularly updated and improved based on user feedback and evolving needs.

Output

The Output of the StayNest project is the fully developed Boarding House Management System, a user-friendly platform designed to streamline the rental process for tenants and property owners. The final system will provide tenants easy access to property listings, booking features, and payment management, improving their overall rental experience. StayNest will automate administrative tasks for property owners, such as listing management, payment tracking, and communication with tenants. The Waterfall methodology ensures that each stage of development builds upon the previous one, resulting in a system that is thoroughly tested, secure, and ready for deployment. By following this structured approach, StayNest will address the key challenges in the boarding house rental process and deliver a solution that enhances operational efficiency for tenants and property owners.

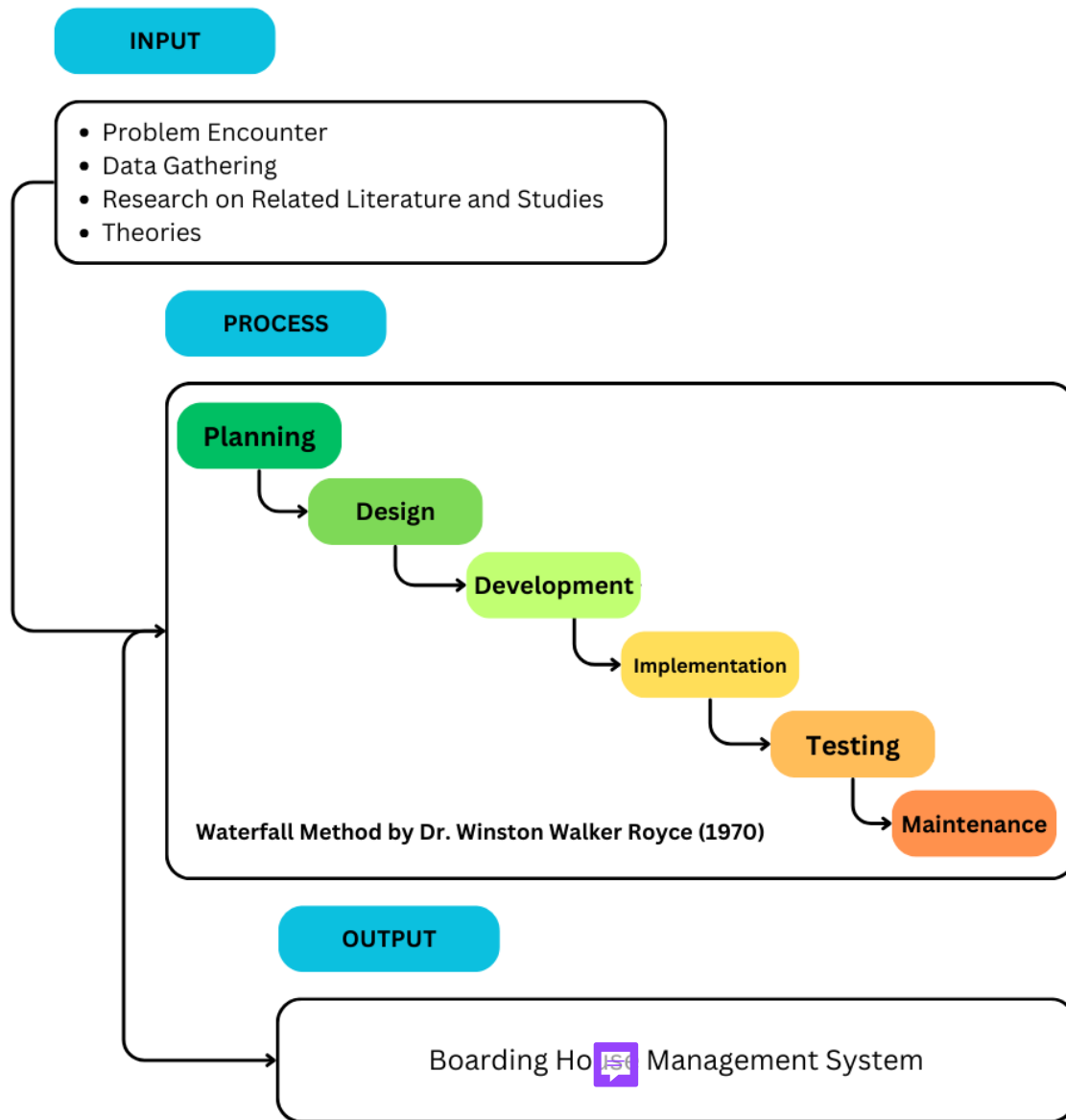


Figure 2. The Conceptual Framework of the Boarding House Management System

CHAPTER IV

METHODOLOGIES, RESULTS, AND DESCRIPTIONS

This chapter presents and explains the various research activities, methods, and findings related to the study.

Methodology

The Board Ease: Boarding House Management and Tenant App was designed, built, and assessed using the Waterfall Method, first proposed by Winston Royce. The project wanted to provide a completely operational platform to enhance the process of discovering and managing boarding house accommodations in Dumaguete City, which is why the Waterfall Method was considered suitable. Planning, Design, Development, Implementation, Testing, and Maintenance define this approach, which guarantees that every phase is finished methodically before advancing to the next. Following this model helped the system development process to be systematic, predictable, and able to efficiently meet user needs and project objectives. The objective of the study was to develop a reliable, user-friendly system that enhances the experience of boarding house owners, caretakers, and tenants.

Several data gathering methods were used to collect the necessary information for the development of the Board Ease: Boarding House Management and Tenant App. To get first-hand understanding about possible challenges, needs, and expectations from a boarding house management system, discussions were held in Dumaguete City with boarding house owners, caretakers, tenants or prospective tenants, and Information Technology consultants. Observations were also carried out to study the current processes used by tenants and owners when searching for or managing boarding house accommodations, allowing the researchers to identify gaps,

inefficiencies, and opportunities for improvement. Furthermore, document analysis was carried out by reviewing existing studies, systems, and related literature to better understand current industry standards, technology developments, and user requirements. This review also aided in identifying successful features employed in similar management systems.

The Waterfall Model's sequential stages were strictly followed during the development of the Board Ease: Boarding House Management and Tenant App. Planning involved identifying problems with existing boarding house management processes and setting objectives for the new system. Analysis focused on gathering and analyzing user needs through interviews and observations. During the Design phase, the structure of the Board Ease: Boarding House Management and Tenant App was developed, including interface layouts, navigation flows, database schemas, and system functionalities. Development involved coding and building the system based on the approved designs, using web and mobile development technologies. Testing evaluated the system's functionality, usability, performance, and security through both internal testing and user acceptance testing. Implementation referred to the deployment of the Board Ease: Boarding House Management and Tenant App among selected boarding house owners, caretakers, and tenants in Dumaguete City. Finally, Maintenance involved regular updates, bug fixes, and improvements based on user feedback and performance monitoring.

Research Environment

The study and development process of Board Ease: Boarding House Management and Tenant App took place in Dumaguete City, Negros Oriental, Philippines, known as the "City of Gentle People." The city has long been recognized as a gateway to Negros Oriental, known for its vibrant culture, historic churches, and growing economy. As a center for education with a large

student population, Dumaguete City faces high demand for boarding houses and rental accommodations. The research focused on observing how property owners manage rentals and how students and workers find rooms, providing the ideal environment to develop and test a more efficient, organized management system. By studying these challenges, the research aims to make the process easier for both tenants and boarding house owners, helping to simplify the search for a good place to stay.

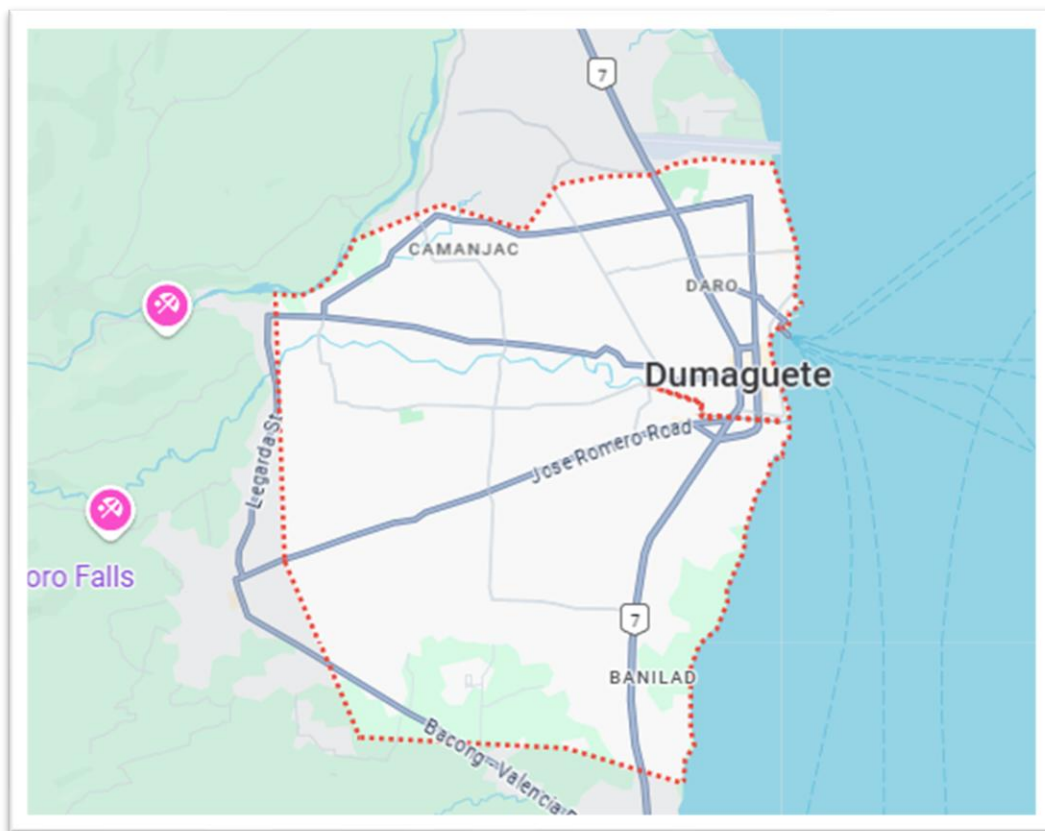


Figure 3. Google Map of Dumaguete City (Research Location)

Population of The Respondents

In this study, a total of one-hundred and fifty (150) respondents were selected to provide relevant feedback and insights regarding the Board Ease: Boarding House Management and

Tenant App. The respondents were composed of ten (10) IT experts who assessed the system's technical performance and usability, nineteen (19) boarding house owners, and ten (10) boarding house caretakers who shared their experiences in managing boarding house operations. Additionally, seventy-one (71) customers, representing individuals actively searching for boarding accommodations, and forty (40) current tenants participated in the study. These groups were carefully chosen because they represent the key users and beneficiaries of the system. Their input played a vital role in shaping the platform to ensure it is practical, user-friendly, and responsive to the real needs of both tenants and property owners.

Instrument

This study's research instrument used a structured rating scale  based on the Technology Acceptance Model (TAM) to assess several elements of the Board Ease: Boarding House Management and Tenant App. The instrument was divided into two primary categories of criteria. The first set evaluated Perceived Usefulness and Perceived Ease-of-Use on a four-point scale, with ratings ranging from 1.00 to 1.75 indicating that the system was "Not Useful" or "Not Easy to Use." Ratings ranging from 1.76 to 2.50 indicated a "Partially Useful" or "Partially Easy to Use" perception. A score of 2.51 to 3.25 classified the system as "Useable" or "Easy to Use," while scores of 3.26 to 4.00 identified it as "Very Useful" or "Very Easy to Use." The second set of criteria evaluated Usability and Sustainability/Maintainability. In this scale, a rating between 1.00 and 1.75 suggested the system was "Not Useable" or "Not Sustainable and Maintainable," 1.76 to 2.50 indicated it was "Slightly Useable" or "Slightly Sustainable and Maintainable," 2.51 to 3.25 reflected a "Moderately Useable" or "Moderately Sustainable and Maintainable" system, and a rating of 3.26 to 4.00 corresponded to being "Very Useful" and "Very Sustainable and Maintainable."

Maintainable." These rating scales provided a clear and measurable framework to analyze participants' perceptions of the system's overall effectiveness, usability, and long-term viability.

Table 1. Rating Scale for Perceived Usefulness and Perceived Ease-of-Use

Rating	Technology Acceptance Model Criteria 	
	Perceived Usefulness	Perceived Ease-of-Use
1.00 – 1.75	Not Useful	Note Easy to Use
1.76 – 2.50	Partially Useful	Partially Easy to Use
2.51 – 3.25	Useable	Easy to Use
3.26 – 4.00	Very Useful	Very Easy to Use

Table 2. Rating Scale for Usability, Sustainability and Maintenance

Rating	Technology Acceptance Model Criteria	
	Usability	Sustainability/Maintainability
1.00 – 1.75	Not Useable	Note Sustainable and Maintainable
1.76 – 2.50	Slightly Useable	Slightly Sustainable and Maintainable
2.51 – 3.25	Moderately Useable	Moderately Sustainable and Maintainable
3.26 – 4.00	Very Useful	Very Sustainable and Maintainable

Statistical Treatment of Data

The following statistical formula in using the treatment of data gathered from the respondents:

1. Percentage

This formula was used in the presentation of the respondents' profile.

$$P = \frac{f}{n} \times 100\%$$

Where:

p - is the percentage

f - is the frequency

n - is the sample size

2. Weighted Mean

This formula was used to analyze the effectiveness of the Boarding House Management System for boarding house owners and tenants.

$$WM = \frac{\sum(W \times X)}{\sum X}$$

Where:

WM - Weighted Mean

W - Weight of each scale

X - Frequency of each scale

Requirements Analysis and Specifications

This part of the study discusses the Board Ease: Boarding House Management and Tenant App's requirements and expected performance. This portion of the research ensures that the system fits the genuine needs of its users, which include both tenants looking for rooms and owners managing their boarding houses. This phase drives the development process and helps avoid future difficulties by precisely outlining the necessary features and how the system should operate.

The Gantt chart illustrates the project timeline for the development of the Board Ease: Boarding House Management and Tenant App covering a period from October 2024 to June 2025. The project is divided into six main activities: Planning, Design, Development, Implementation, Testing, and Maintenance. Planning is scheduled to take place from mid-October to early November 2024, laying the groundwork for the subsequent phases. The Design phase begins in November 2024 and extends through March 2025, focusing on structuring the system and outlining its technical specifications. Development is set to start in December 2024 and continues until mid-May 2025, involving both frontend and backend coding, as well as database integration. Implementation occurs from April to mid-May 2025, where the system is deployed, servers are configured, and the platform becomes operational. Testing is conducted in May 2025, ensuring the system's usability, functionality, and performance through rigorous assessments. Finally, Maintenance is scheduled for June 2025 to address any post-deployment issues, perform updates, and ensure long-term system stability. This timeline reflects a well-organized, phased approach to system development, allowing for thorough planning, execution, and refinement.

Gantt Chart

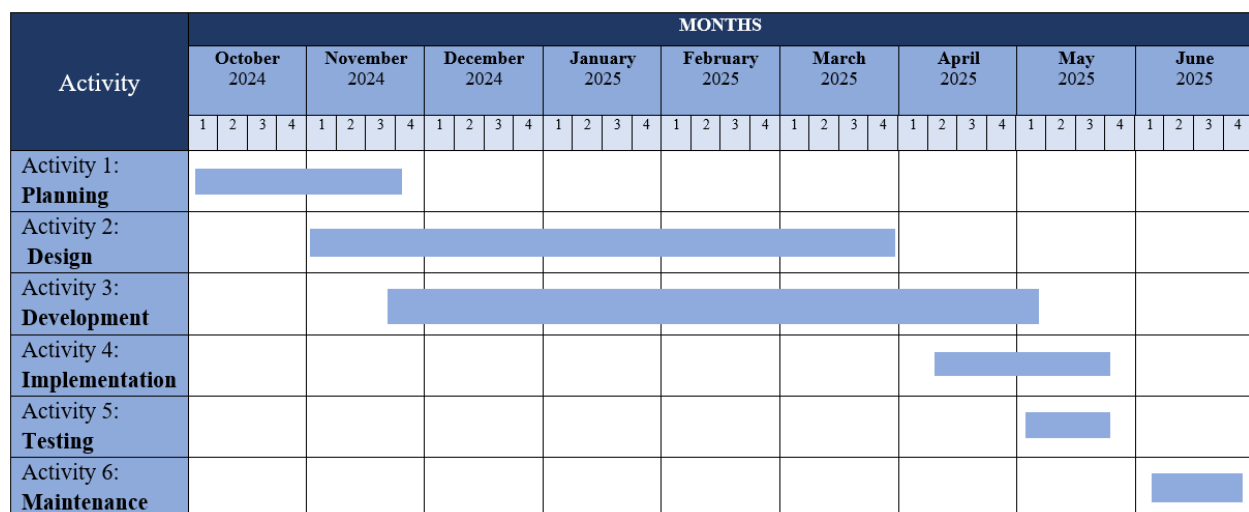


Figure 4. Gantt Chart

Requirements Modeling

This section of the study presents the requirements modeling, which aims to systematically define the functional and non-functional requirements of the system based on user needs and existing challenges. It serves as a critical foundation for developing a structured solution that addresses the inefficiencies in boarding house management.

Input

The system begins collecting input when a user accesses the platform. Users may enter search criteria such as location, rental budget, and desired amenities, which the system uses to generate a list of available boarding houses. When a user selects a preferred property and chooses to book or schedule a visit, they are required to either log in or create an account. For login, users must provide a valid email address and password. New users must complete a registration form by submitting their full name, email address, password, and uploading a valid identification document. Once authenticated, the user submits their preferred date and time for a site visit. The system stores this schedule and notifies the property owner for confirmation. After the site visit, if the user decides to reserve a room, the owner must approve the reservation before proceeding to payment. At this stage, users choose a payment method—either GCash through the Staynest application or cash directly to the owner. For GCash, the system collects the user's mobile number and payment reference. For cash transactions, the user is instructed to pay the owner during the visit or upon moving in.

Process

Initially, the system processes the user's search input and displays a list of boarding houses that meet the criteria. When a visit request is made, the system verifies whether the user is logged in. If not, the user is directed to either log in or register. Once verified, the system saves the visit request and alerts the property owner for review. After the site visit, the user may proceed to submit a reservation request. This action also requires approval from the property owner. Upon approval, the system activates the payment process. If the user selects GCash, the system verifies the transaction and stores the reference number. For cash payments, the property owner manually confirms receipt using the system's administrative dashboard. After successful payment, the user's status is updated to "Tenant," and the room's availability is accordingly modified.

Output

The initial output is a tailored list of available boarding houses based on the user's input. After a visit request is submitted, the user receives a notification indicating that the request is awaiting owner approval. Once the owner responds, the system notifies the user of the outcome. If the reservation is approved and payment is completed, the system displays a confirmation message along with a digital receipt or transaction reference number. The user's role changes to "Tenant," and their dashboard is updated with tenant-specific features such as maintenance request options and payment history. Concurrently, the property owner is provided with real-time updates on visit requests, reservation decisions, and tenant information, enabling efficient and organized property management.

Feedback

Feedback plays a vital role in the continuous improvement and effectiveness of the Board Ease: Boarding House Management and Tenant App. It serves as a valuable mechanism for gathering insights from both tenants and property owners regarding their experiences with the platform. Through structured feedback channels, users are able to share comments, suggestions, and concerns about various features, such as the booking process, payment options, user interface, and overall system reliability. Property owners, for instance, may provide input on how efficiently the system notifies them of booking requests and tenant activities, while tenants might comment on the ease of navigating listings or the clarity of payment procedures. This feedback is systematically reviewed by the development team to identify recurring issues and prioritize enhancements. By incorporating user feedback into system updates, Board Ease ensures that the platform remains responsive to user needs, ultimately leading to higher satisfaction, increased trust, and better service delivery. In this way, feedback is not only a tool for assessment but also a driver of innovation and sustained system relevance.

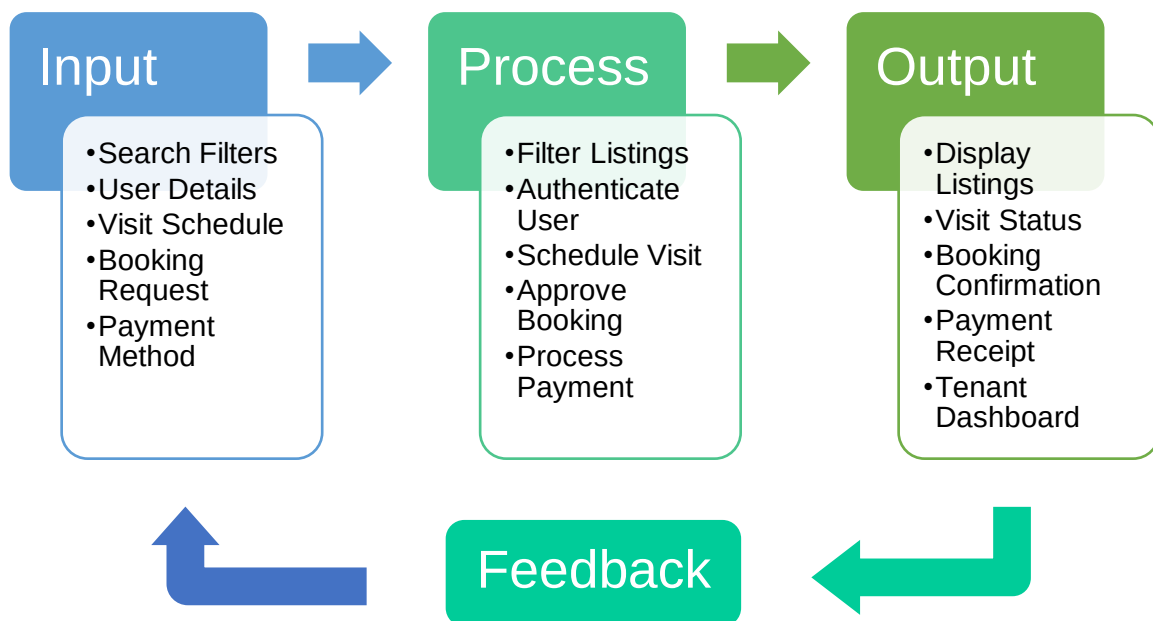


Figure 5. IPO Process Modeling

Data and Process Modeling

This portion of the study focuses on Data and Process Modeling, which outlines the flow of information and the interactions between different components of the StayNest Boarding House Management System. It helps visualize how data is processed, stored, and transferred within the system, ensuring that all operations are clearly defined and aligned with user requirements.

Data Modeling

Data modeling involves identifying and organizing the data elements essential to the Board Ease: Boarding House Management System. It defines the structure of the system's data, including entities, attributes, and their relationships, to ensure accurate data storage, retrieval, and integrity. This model serves as a blueprint for designing a reliable and efficient database that supports the system's overall functionality.

Entity Relationship Diagram

This section presents the Entity Relationship Diagram (ERD) of the system, illustrating the logical structure of the database and the relationships among its entities. It provides a clear visual representation of how key data elements—such as users, properties, bookings, and payments—are interconnected. The ERD serves as a foundational tool in the system's design, ensuring data consistency, integrity, and efficient organization throughout the application.

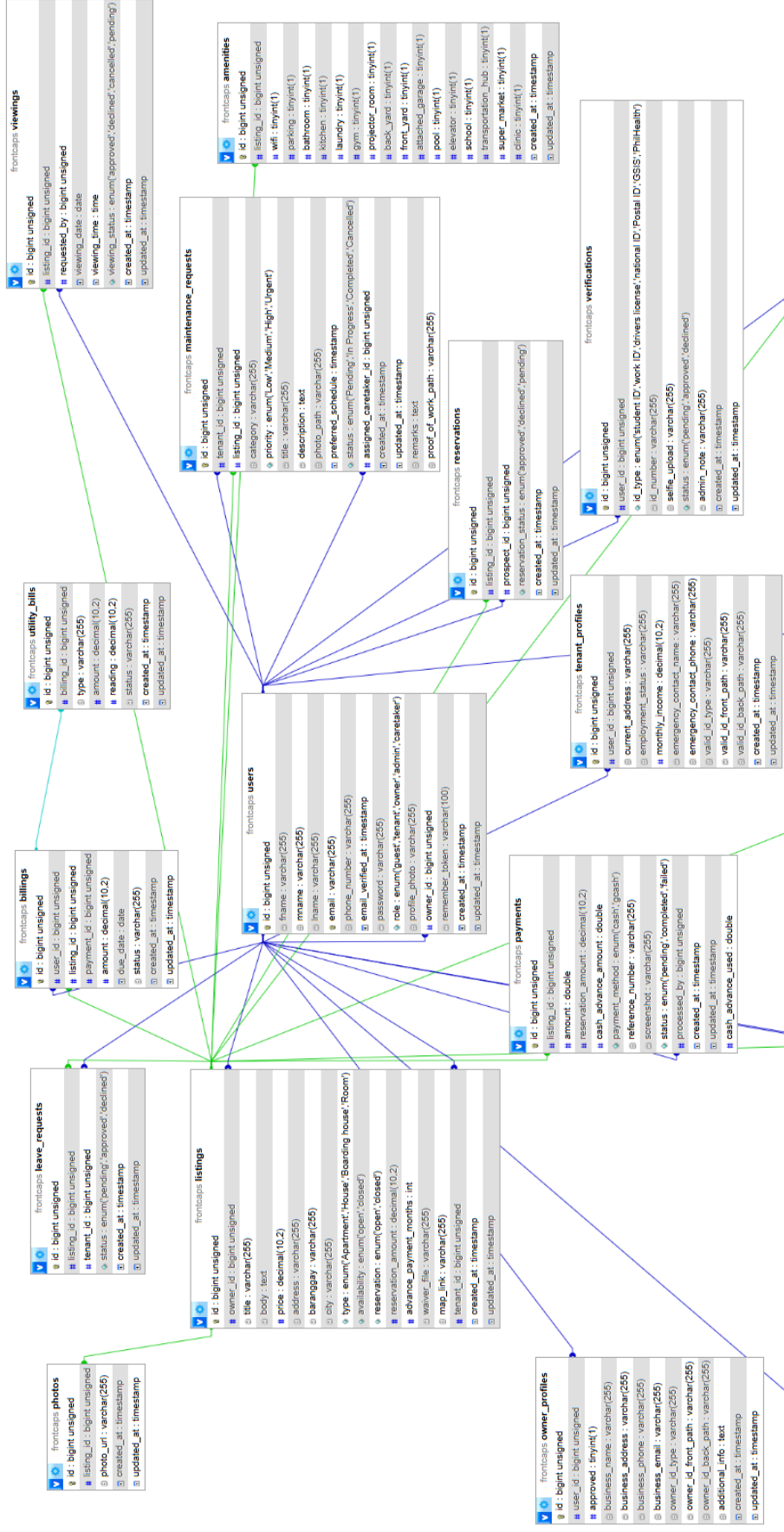


Figure 6. Entity Relationship Diagram

The context diagram for the Board Ease: Boarding House Management and Tenant App illustrates the interaction between the system and its main external entities: Tenants/Prospective Tenants, Administrators, and the Payment Gateway (GCash). Tenants or prospective tenants interact with the system by registering or logging in, searching for available boarding houses, booking rooms, and communicating with owners or administrators. These actions are submitted as requests to the system, which responds with appropriate services such as listings, booking confirmations, and messages. Administrators perform two major roles within the system: they manage users by verifying listings and addressing complaints, and they oversee property-related functions such as registering owners, managing listings, and approving or declining booking requests. The system handles these requests and provides timely responses to maintain smooth platform operations. Additionally, the integration with the Payment Gateway (GCash) enables tenants to make online payments. Once a payment request is initiated, the system communicates with GCash, which processes the transaction and sends back a confirmation that is then reflected in the app. This structured flow of data ensures that BoardEase delivers a seamless and efficient experience for all its users, enhancing the management and rental of boarding houses.

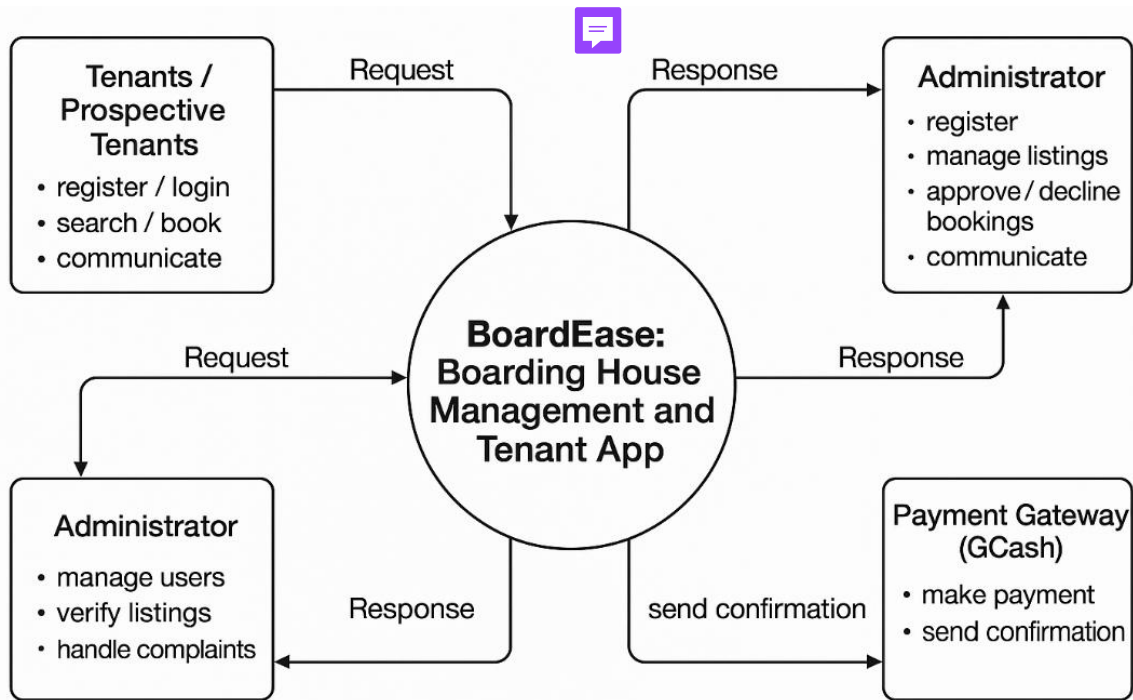


Figure 7. Context Diagram

This data flow diagram illustrates a boarding house management system where users can search for available boarding houses, log in or sign up, and book visits, with booking details being stored in the Bookings DB. If not logged in, users are directed through a login/signup process that validates their data in the Users DB. Upon booking, owners receive notifications and can approve or decline bookings, triggering payment initialization and updates in the Payments DB. Users then process payments and receive confirmation. Users can also request maintenance, which is logged in the Maintenance Requests DB and alerts the caretaker. Admins manage the system by approving owner accounts, creating caretaker accounts, and updating user and property data in the Users DB and Boarding Houses DB. This centralized system facilitates seamless booking, payments, maintenance, and account management among users, owners, caretakers, and admins.

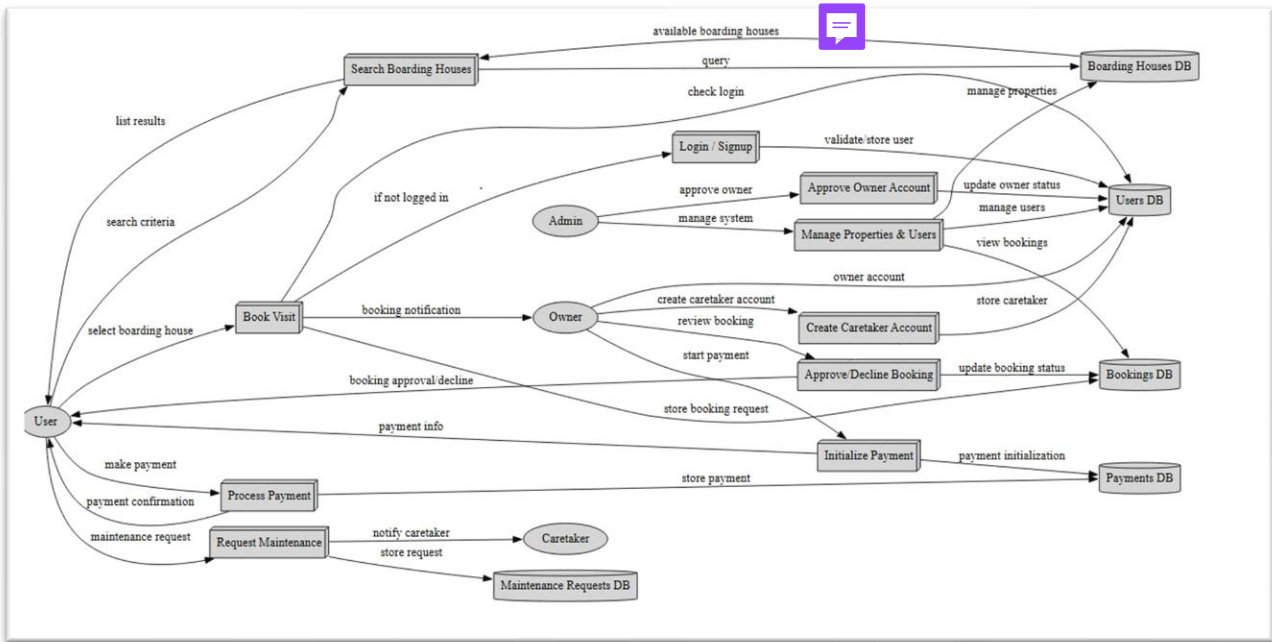


Figure 8. Data Flow Diagram

Object Modeling

Figure 9 illustrates a use case diagram for a Boarding House Management and Tenant App, showing the interactions between three main user roles—Prospective Tenants, Caretakers, and Owners—and the system’s functionalities. Prospective Tenants can register, log in, search content, browse or create listings, make and manage bookings or reservations, transact payments, and submit maintenance requests. Caretakers are involved in logging in, handling maintenance requests, and managing leave requests. Owners have the most extensive access, including registering, logging in, searching content, creating listings, managing bookings and reservations, handling payments, responding to maintenance and leave requests, managing billing and property details, and creating staff accounts. The diagram demonstrates how the system supports each role with tailored functions, ensuring a streamlined rental and property management experience.

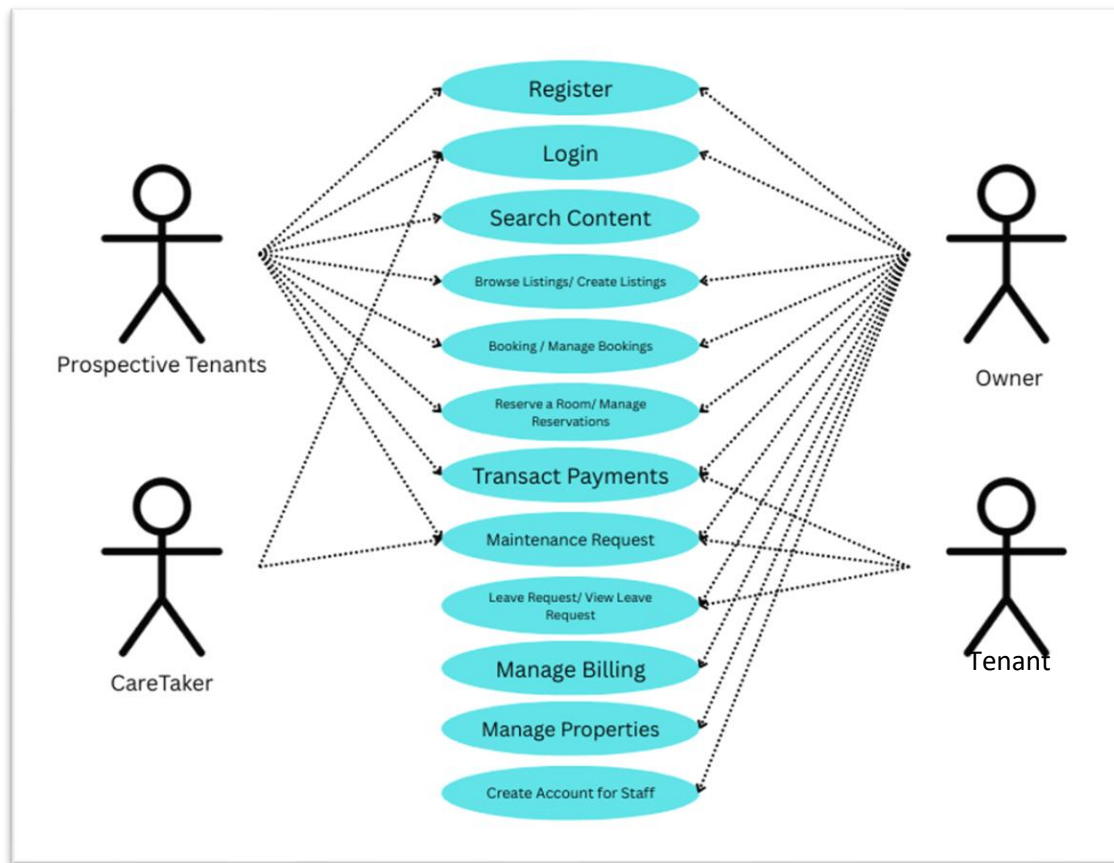


Figure 9. Use Case Diagram

Requirements Documentation

This section provides a comprehensive overview and detailed descriptions of the documented requirements necessary for the development and implementation of the StayNest Boarding House Management System. It outlines the functional and non-functional requirements gathered from stakeholders to ensure that the system meets user expectations and operates effectively throughout its intended use.

User-Interface Design and Reports

This section provides a concise overview of the system's user-interface design and reporting features. It highlights the visual structure, interactive components, and layout choices that contribute to a user-friendly experience. Additionally, it presents the reporting functionalities, showcasing how data is organized and displayed to support efficient decision-making for tenants, boarding house owners, and caretakers. The interface was carefully designed with accessibility and ease of navigation in mind, ensuring that users of all backgrounds can interact with the system effortlessly. The interface and reporting tools enhance usability, streamline management tasks, and promote transparency across the system.

Log-in Page

The login page serves as the secure access point for users who already have an existing Staynest account. The interface consists of a clean and minimalist layout that prompts the user to enter their registered email address and corresponding password. These input fields are centrally aligned to ensure ease of use and visual clarity. Below the login form, there is an option for users who may have forgotten their password, directing them to a recovery process. The page maintains a consistent design theme with the rest of the system, incorporating subdued color schemes, legible typography, and responsive elements to enhance the user experience across devices. Additionally, subtle feedback messages guide users through errors, such as incorrect credentials, enhancing usability and reducing frustration. Overall, the page balances functionality with aesthetic simplicity to provide a smooth and reassuring entry point into the StayNest platform.

STAYNEST Home Listings Login Sign Up + Add listing

Please log in with your account!

Login to your account easily with less information.

Email address *

Password *

Your password must be 8 characters at least.

☐ Keep me signed in

[Forgot password?](#)

[Sign in](#)

Don't have an account? [Signup here](#)

Figure 10. Login page

Sign-up Page

Figure 11 illustrates the first segment of the two-step sign-up form designed for new user registration within the Board Ease: Boarding House Management and Tenant App. This interface gathers essential personal details, including first name, middle name, last name, email address, mobile number, and the user's chosen password. Each input field is clearly labeled and equipped with built-in validation features that proactively help users avoid mistakes and ensure the accuracy of submitted data. After completing the required information, users can proceed by clicking the "Next Stage" button positioned at the bottom of the form. The layout is thoughtfully designed to be intuitive, guiding users smoothly through the registration process step-by-step, which effectively minimizes the chances of incomplete or incorrect submissions. This initial stage serves both tenants and boarding house owners alike, establishing a secure and reliable foundation of user identity within the system and ensuring a consistent onboarding experience for all.

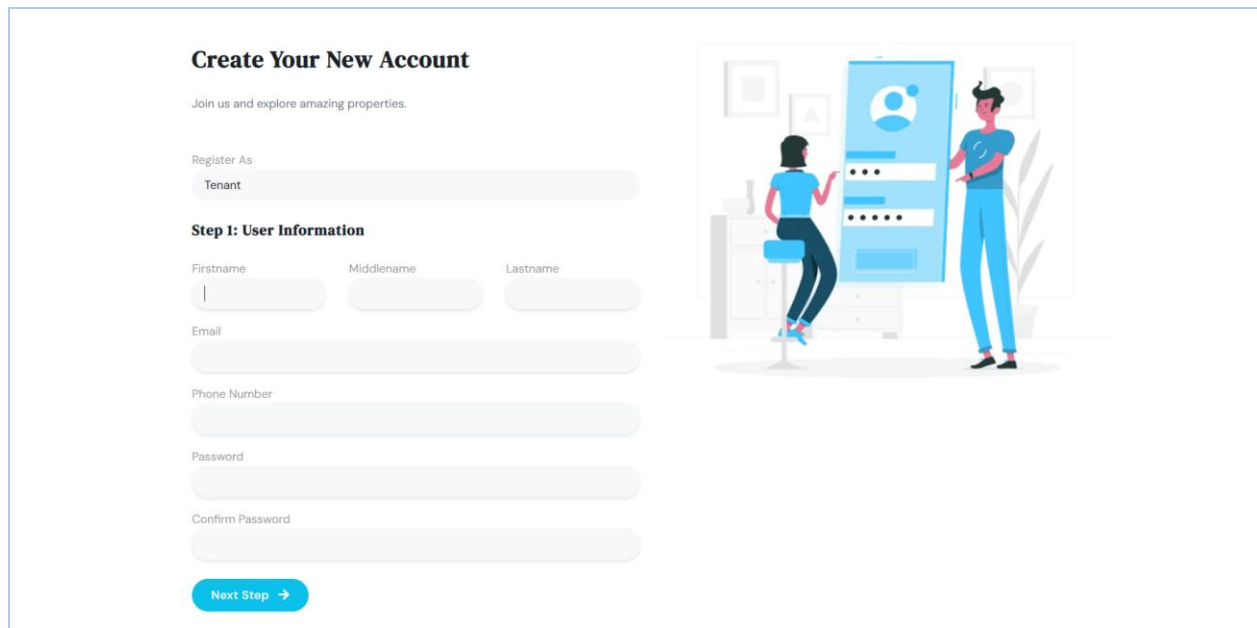


Figure 11. First part of Sign-up page

Figure 12 continues the user registration process by displaying the second part of the sign-up form, which collects more detailed personal information crucial for building verified profiles within the Staynest system. This section prompts users to provide their current address, employment status, estimated monthly income, and emergency contact details, offering a fuller picture of each individual's background and ensuring preparedness for unexpected situations. An integrated file upload tool allows users to easily submit a valid identification document, a vital step that supports the platform's robust security and identity verification protocols. By requiring this additional layer of information and documentation, StayNest fosters greater trust and accountability among tenants and boarding house owners alike. Once all fields are completed and the ID is uploaded, users can click the prominently placed "Register" button to finalize their registration, effectively securing their place within the system and enabling access to its full range of features.

Create Your New Account

Join us and explore amazing properties.

Register As

Tenant

Step 2: Profile Information

Current Address

Employment Status

Select Employment Status

Monthly Income

Emergency Contact Name

Emergency Contact Phone

Valid ID Type

Select ID Type

Upload Valid ID Front

Choose File | No file chosen

Upload Valid ID Back

Choose File | No file chosen

Previous Step ←

Register ✓

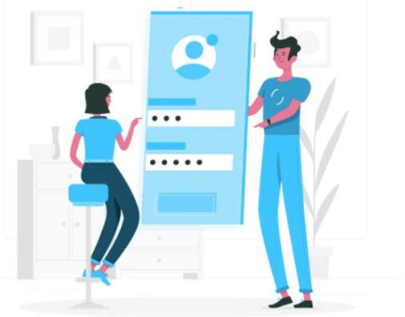


Figure 12. Second part of Sign-up page

Owner Dashboard

The boarding house owner’s dashboard serves as the central hub for managing and overseeing all key aspects of their operations within the Staynest platform. It features a comprehensive overview panel that presents critical metrics at a glance, including the total number of properties currently posted for listings, pending maintenance requests, the number of booking inquiries from prospective tenants, and overall payment summaries. In addition, the dashboard includes convenient quick-access widgets that highlight recent activities, upcoming tasks, and detailed payment analytics, such as monthly revenue trends and the total number of payments processed. Designed with a clean, well-structured layout and an intuitive arrangement of visual components, the dashboard empowers owners to efficiently monitor their business performance, prioritize tasks, and make informed decisions with ease.

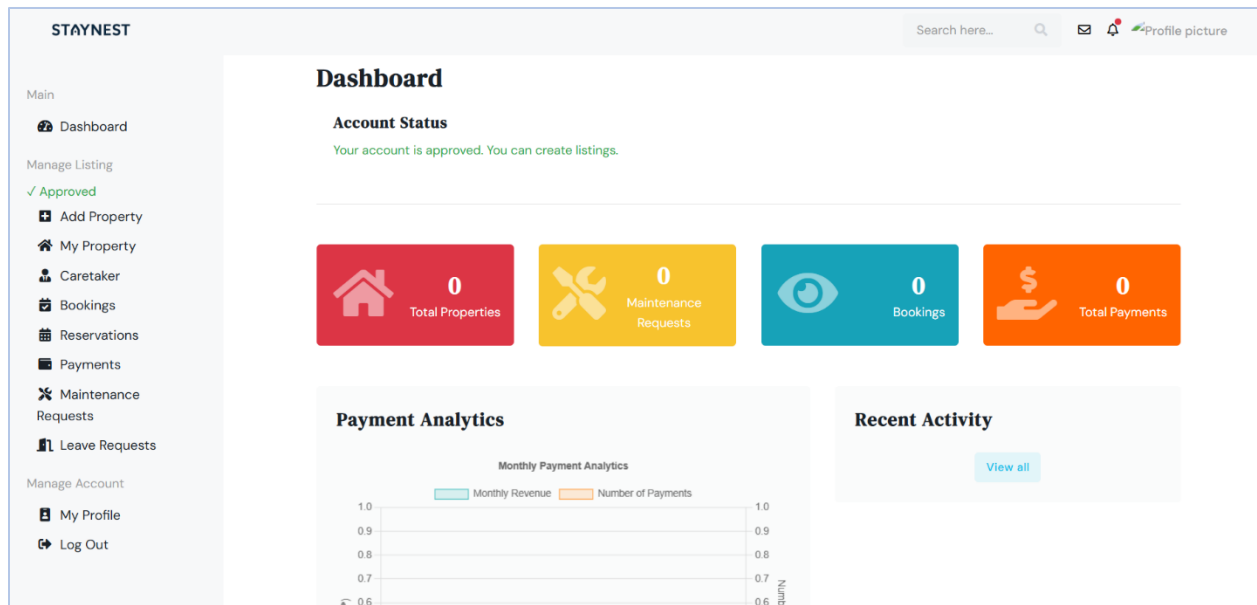


Figure 13. Owner Dashboard

Navigation Bar of Owner Side

The navigation bar on the owner's side functions as the primary navigation tool, offering structured and intuitive access to all essential modules within the system. Positioned vertically along the side of the screen, the navigation bar is designed to remain fixed during user interaction, ensuring uninterrupted access to key features as users move between pages. It includes clearly labeled links to major sections such as Dashboard, My Property, Add Property, Caretaker Management, Bookings, Reservations, Payments, Maintenance, Leave Requests, and Profile. Each label is paired with a corresponding icon to enhance visual recognition and improve navigation efficiency, especially for users managing multiple properties or tasks at once. The layout is thoughtfully organized to prioritize accessibility and reduce cognitive load, allowing boarding house owners to quickly find and utilize system functions without unnecessary complexity.

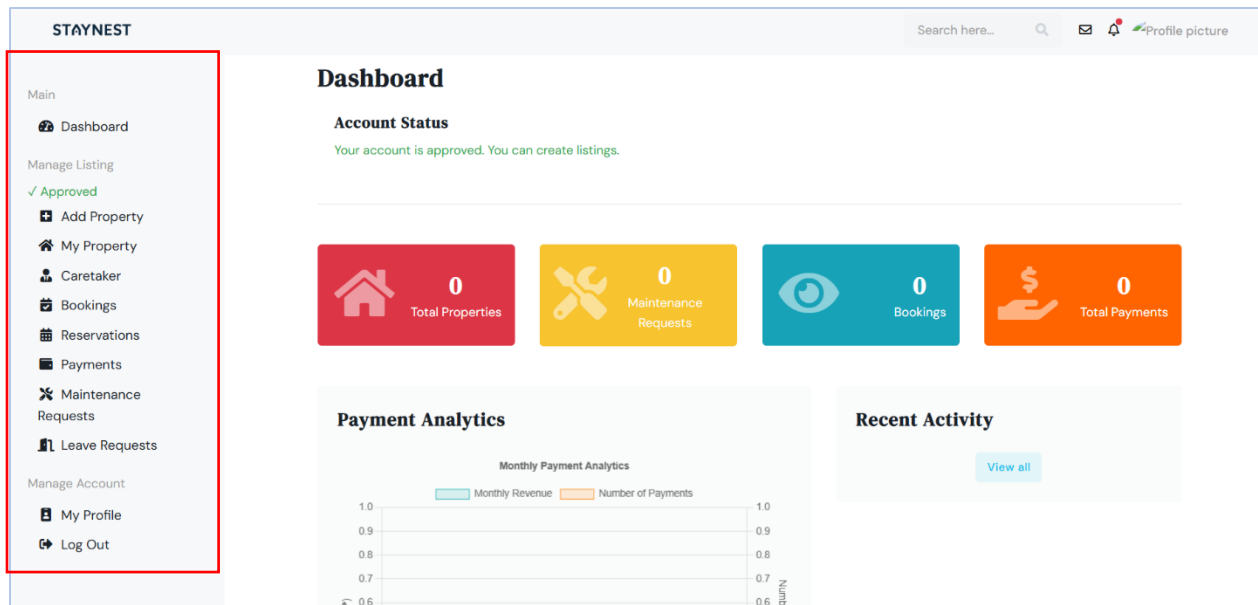


Figure 14. Navigation bar of owner side

Add Property

Figure 15 illustrates the initial step in the Add Property process, which centers on entering the fundamental details of the boarding house to establish its identity within the Staynest system. This section features a set of clearly labeled input fields designed to capture essential property information, including the property title, a concise yet descriptive overview, the barangay and city location, as well as the complete address—comprising the street name, building name, and unit or block number. This foundational step is crucial for ensuring that each property is accurately documented and easily searchable. The layout is clean and intuitively organized, with ample spacing and instructional placeholders that guide the owner through the data entry process efficiently and with minimal confusion. By collecting precise location details and a meaningful description at this early stage, the system lays the groundwork for effective property listings and

enhanced visibility to potential tenants. Overall, this step streamlines the process of onboarding new properties, setting a professional tone for the rest of the property management workflow.

The screenshot shows the STAYNEST web application interface. The main heading is "Find Your Perfect Tenant" in large blue font, with the subtitle "Create your property listing in just a few simple steps". Below this is a progress bar with four steps: "Basic info" (selected), "Details", "Amenities", and "Photos". The "Basic info" step is expanded into a form titled "Basic Information". The form contains the following fields: "Title" (with a placeholder "e.g. Spacious 2-Bedroom Apartment near City Center"), "Description" (with a placeholder "Describe your property's features, neighborhood, and what makes it special..."), "City" (with a location pin icon and the text "Barangay"), and "Complete Address" (with a placeholder "Full address including street number, building name, etc."). A blue "Continue" button with a right arrow is at the bottom right of the form. On the left side of the screen is a sidebar menu with options: "Main" (Dashboard), "Manage Listing" (Add Property, My Property, Caretaker, Bookings, Reservations, Payments, Maintenance Requests, Leave Requests), and "Manage Account" (My Profile, Log Out). At the bottom of the form, there is a link "Tips for a Great Listing".

Figure 15. Add Property

Figure 16 presents the second step of the Add Property form, where boarding house owners are prompted to provide detailed listing information that defines the core attributes of their rental offering. In this section, owners specify the type of accommodation available such as a boarding house, apartment, private room, or shared space, allowing the system to categorize listings accurately and assist tenants in refining their search preferences. Additional fields capture the monthly rental rate and the required reservation fee, ensuring that prospective tenants are aware of the financial commitments involved upfront. Owners are also asked to include a Google Maps link to pinpoint the exact location of the property, enhancing location transparency and helping users assess accessibility. Furthermore, a waiver or special terms related to occupancy can be uploaded or noted, allowing owners to clarify important conditions or agreements. A toggle feature is

included to indicate whether the listing is currently available and open for reservations, which helps maintain up-to-date and reliable property information. To support ease of use, this step incorporates dropdown menus, numeric input fields, and clearly labeled sections that streamline the data entry process while minimizing the risk of input errors.

The screenshot shows the 'Find Your Perfect Tenant' interface on the STAYNEST platform. The main heading is 'Find Your Perfect Tenant' in blue, with a subtext 'Create your property listing in just a few simple steps'. A progress bar at the top indicates the current step is 'Details', with 'Basic Info' as the previous step and 'Amenities' and 'Photos' as future steps. The 'Listing Details' form is divided into several sections: 'Property Type' with a dropdown menu; 'Advance Payment' with a dropdown menu set to 'No advance payment'; 'Availability' with two buttons, 'Available Now' (green) and 'Not Available' (red); 'Reservation' with two buttons, 'Accepting Reservations' (green) and 'No Reservations' (red); 'Monthly Price' with a numeric input field set to '0.00'; 'Reservation Fee' with a numeric input field set to '0.00'; 'Google Maps Link' with a text input field containing 'https://maps.google.com/'; and 'Waiver File' with a 'Choose File' button and a 'No file chosen' status. A 'Continue' button is at the bottom right. Below the form, there are 'Tips for a Great Listing' including 'Be Descriptive', 'Quality Photos', and 'Be Honest'.

Figure 16. Add Property

Figure 17 displays the third step in the property listing process, where boarding house owners are prompted to specify the amenities and features available within their property. This section includes a user-friendly checklist of commonly sought-after conveniences such as Wi-Fi, air conditioning, kitchen access, laundry facilities, parking space, and more. Owners can select multiple options to provide a clear and comprehensive overview of what the property offers, ensuring that potential tenants have a full understanding of the comfort and functionality available to them. This step plays a vital role in enhancing listing accuracy and overall transparency, as it allows prospective tenants to quickly filter and compare properties based on their personal

preferences, lifestyle requirements, and non-negotiable daily needs. By presenting this information in a structured and visually accessible format, the system simplifies decision-making for tenants and increases the chances of successful, satisfaction-driven bookings. In turn, this improves user trust and reduces the likelihood of mismatched expectations upon move-in.

The screenshot displays the 'STAYNEST' web application interface for adding a property listing. The main heading is 'Find Your Perfect Tenant' with the subtext 'Create your property listing in just a few simple steps'. A progress bar indicates the current step is 'Amenities'. The 'Amenities' section asks the user to 'Select all the amenities that your property offers:' and provides a grid of 15 amenity options, each with a checkbox and an icon. The options are: Wi-Fi, Kitchen, Projector Room, Attached Garage, School, Clinic, Parking, Laundry, Back Yard, Pool, Transportation Hub, Bathroom, Gym, Front Yard, Elevator, and Super Market. The 'Continue' button is highlighted in blue. A sidebar on the left contains navigation links for Main, Manage Listing, and Manage Account. A search bar and profile picture icon are in the top right corner.

Figure 17. Add Property

Figure 18 shows the final step in the Add Property process, which focuses on uploading images of the boarding house to visually showcase the space to potential tenants. This section features a convenient drag-and-drop upload interface, allowing owners to effortlessly attach multiple high-quality photos of the property. Recommended images include shots of individual rooms, shared common areas such as kitchens and living rooms, bathrooms, and exterior views of the building and its surroundings. These visual elements are essential in increasing the attractiveness, transparency, and credibility of the listing, as they offer prospective tenants a more accurate and immersive understanding of the property layout and condition—often influencing

their decision to inquire or reserve. By integrating visual content in the final step, StayNest empowers property owners to present their spaces in the best possible light, while also helping tenants make more confident and informed choices without needing to rely solely on text-based descriptions.

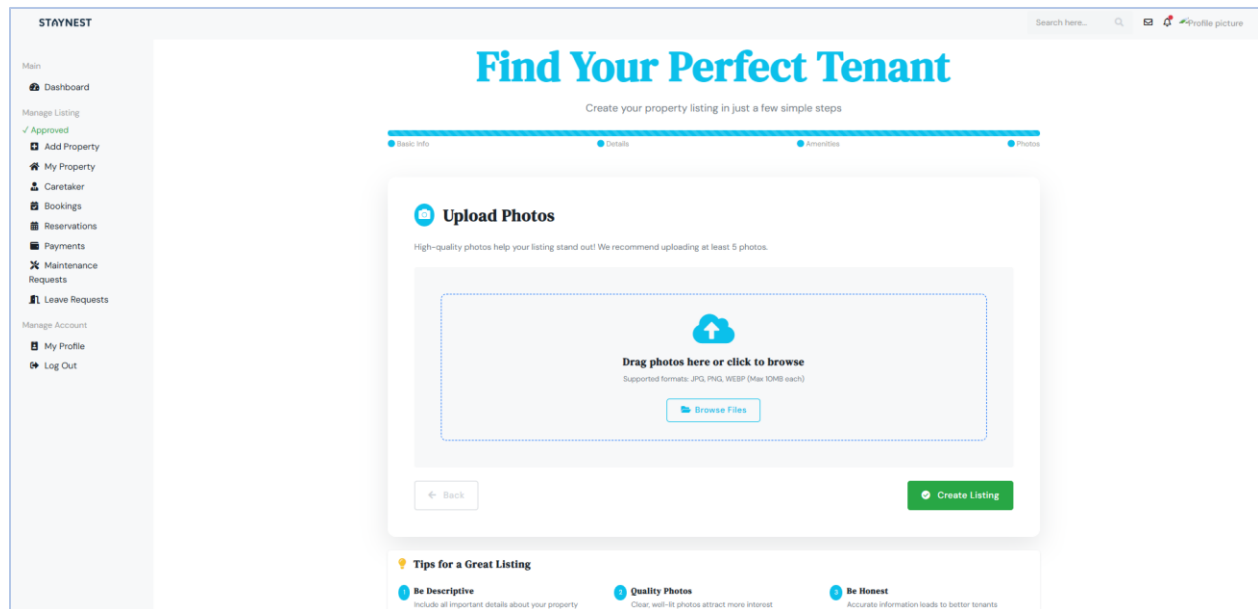


Figure 18. Add Property

My Property

This page provides owners with a detailed list of all their registered boarding house listings. Each property is displayed in a card-style layout that includes the property title, thumbnail image, current status (active, pending, or archived), and a summary of key details such as rental rate, number of rooms available, and location. A set of action buttons—such as “Edit,” “View,” or “Remove”—allows owners to make real-time updates to each listing. The page also incorporates a search bar and filtering tools to help users quickly locate specific properties based on status or

location. This feature-rich interface ensures efficient property management and enhances organizational control.

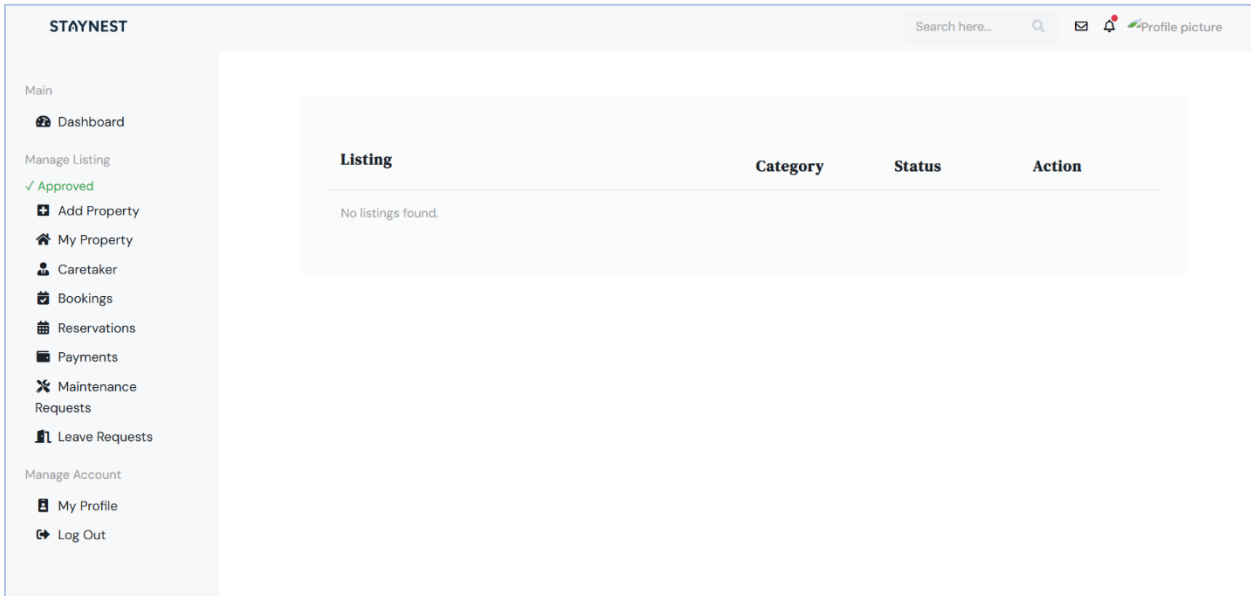


Figure 19. My Property

Caretaker (Owner Side)

Figure 20 illustrates the caretaker page in the owner’s side. The caretaker management page enables property owners to efficiently assign, register, and oversee caretaker roles for their listed properties, streamlining the delegation of operational responsibilities. Through this interface, owners can add new caretakers by creating individual accounts for them—inputting key personal details such as full name, contact number, email address, and a secure password for system login. This allows caretakers to access their own dedicated dashboard and manage tasks specific to the properties assigned to them. The page features a well-organized table that lists all active caretakers, with clearly defined columns displaying their name, email, and contact information. Each row is equipped with intuitive action buttons that allow owners to edit caretaker details or remove

caretakers as needed. This centralized and user-friendly module promotes clear communication and task delegation, ensuring that day-to-day responsibilities such as property maintenance, tenant inquiries, and routine inspections are effectively handled by trusted personnel.

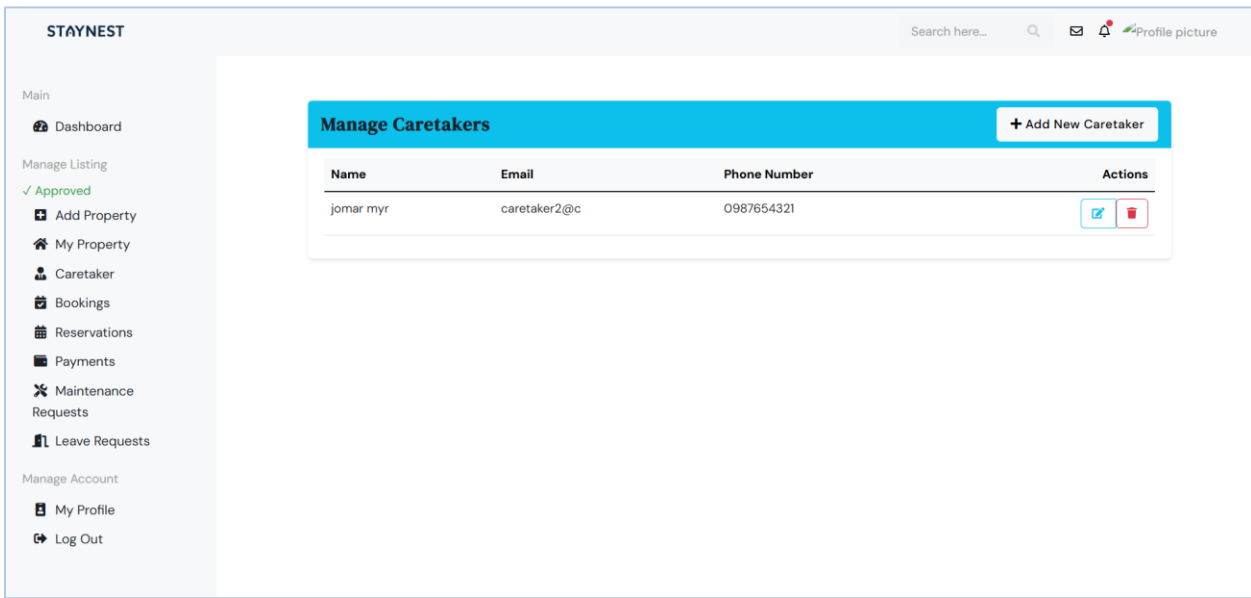
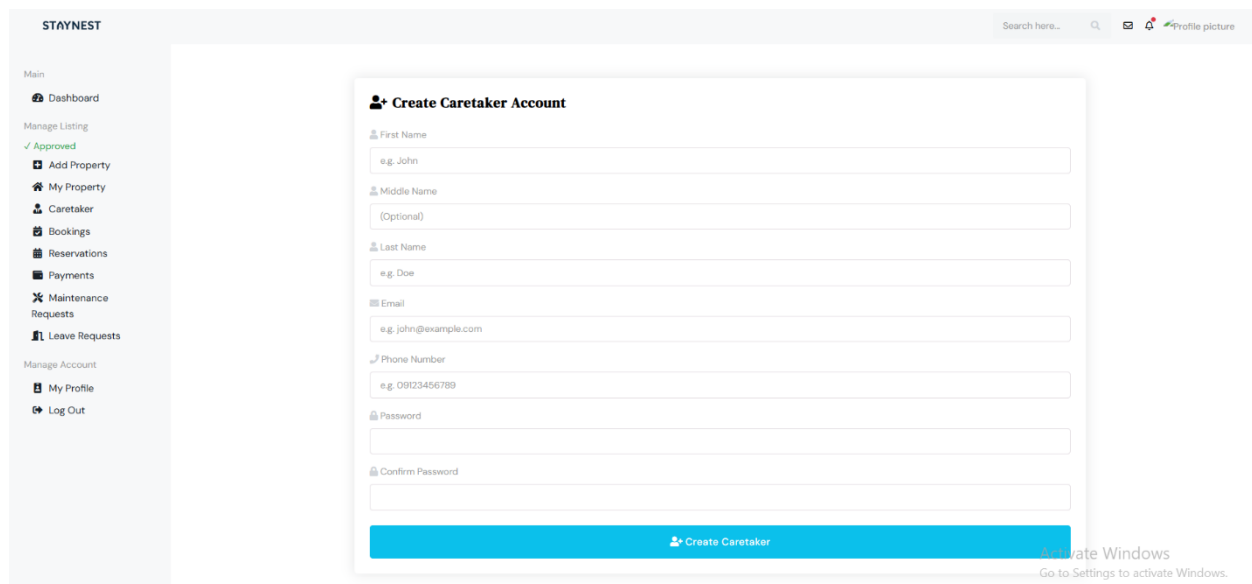


Figure 20. Caretaker (Owner Side)

Add Caretaker

Figure 21 illustrates the Create Caretaker Account interface, which allows property owners to easily register and set up system access for their designated caretakers. This page is designed to be straightforward and user-friendly, guiding the owner through the process of creating a new caretaker account. To complete the registration, the owner is required to input key information, including the caretaker’s full name, a valid email address, phone number, and a secure password that will be used for logging into the system. These details ensure that each caretaker has unique login credentials and can access only the tools and features relevant to their assigned responsibilities. Once all required fields have been filled out, the owner can simply

click the "Create Caretaker" button to finalize the process. This step is essential in granting caretakers the ability to manage daily property-related tasks, communicate with tenants, and respond to maintenance requests through their own dashboard. The intuitive design and clear field labels make the process efficient, while also helping to maintain secure and organized personnel records within the Board Ease platform.



The screenshot displays the 'Create Caretaker Account' form within the STAYNEST application. The interface includes a top navigation bar with the 'STAYNEST' logo, a search bar, and a profile picture icon. A left sidebar menu lists various navigation options: Main (Dashboard), Manage Listing (Add Property, My Property, Caretaker, Bookings, Reservations, Payments, Maintenance Requests, Leave Requests), and Manage Account (My Profile, Log Out). The main content area features the 'Create Caretaker Account' form, which contains the following fields: First Name (with placeholder 'e.g. John'), Middle Name (with placeholder '(Optional)'), Last Name (with placeholder 'e.g. Doe'), Email (with placeholder 'e.g. john@example.com'), Phone Number (with placeholder 'e.g. 09123456789'), Password, and Confirm Password. A blue 'Create Caretaker' button is positioned at the bottom of the form. A Windows watermark is visible in the bottom right corner of the image.

Figure 21. Add Caretaker

Bookings

The booking page displays all booking requests submitted by prospective tenants. Each request is organized in a tabular format or card view, showing essential details such as the name of the tenant, requested room, booking date, and status (pending, approved, or declined). Property owners have access to review each booking in detail, which includes tenant profile information, visit history, and intended length of stay. Action buttons allow the owner to approve or reject the request, with automated notifications sent to the tenant accordingly. This page provides

transparency and control over tenant intake, ensuring that room reservations are processed efficiently.

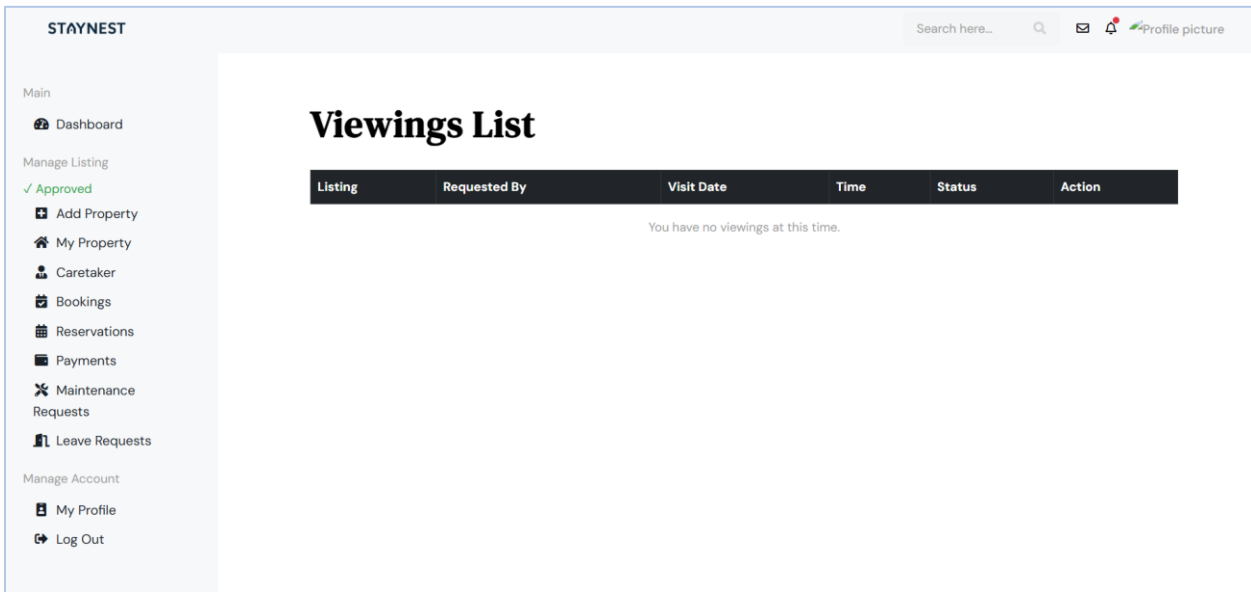


Figure 22. Bookings

Reservations

The reservation page consolidates all approved bookings into a single interface, offering owners an overview of room occupancy. Each reservation entry includes the tenant’s name, property address, check-in and check-out dates, payment status, and contact information. A visual status bar may be included to indicate progress from reservation to tenancy. Owners can also view any upcoming move-ins and make necessary adjustments to room assignments. This page plays a crucial role in maintaining accurate tenant records and planning future occupancy.

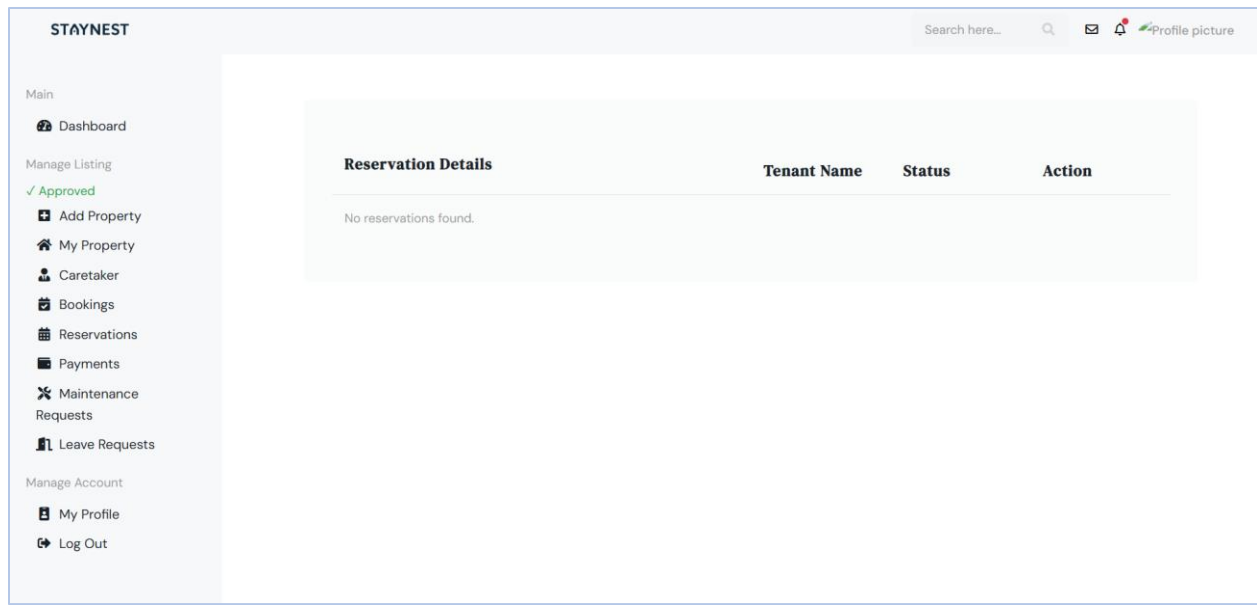


Figure 23. Reservations

Payments

The payment page serves as a financial tracking tool that records all transactions related to room bookings. It displays information such as the name of the tenant, payment amount, date of transaction, chosen payment method (GCash or cash), and transaction reference number. For GCash payments, the system automatically verifies and stores reference codes. For cash payments, the property owner manually confirms receipt via the system dashboard. The page features a filter for sorting by date, payment status, or property, and provides downloadable records to support accounting needs and audit trails.

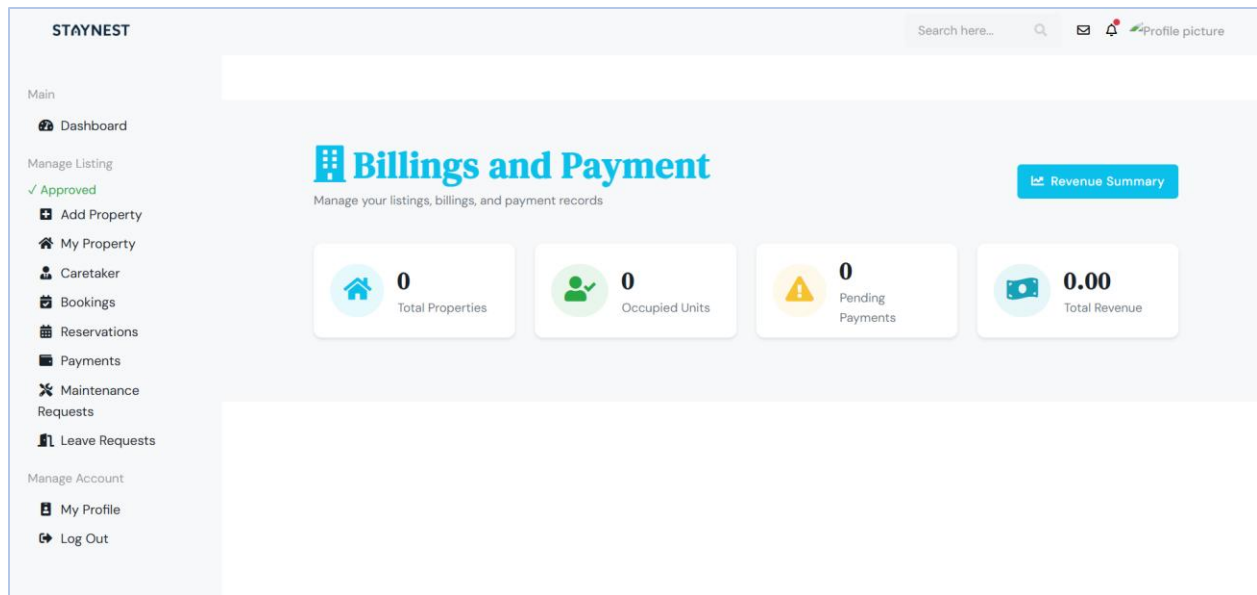


Figure 24. Payments

Maintenance Requests

The maintenance page centralizes all tenant-submitted requests for repairs or service. Each request is listed with details such as the issue type (e.g., plumbing, electrical), submission date, tenant name, and assigned caretaker. Owners can update the status of each request (e.g., pending, in progress, resolved), and include notes or follow-up dates as necessary. The interface allows for sorting and filtering based on urgency or property, supporting a responsive and accountable approach to property upkeep.

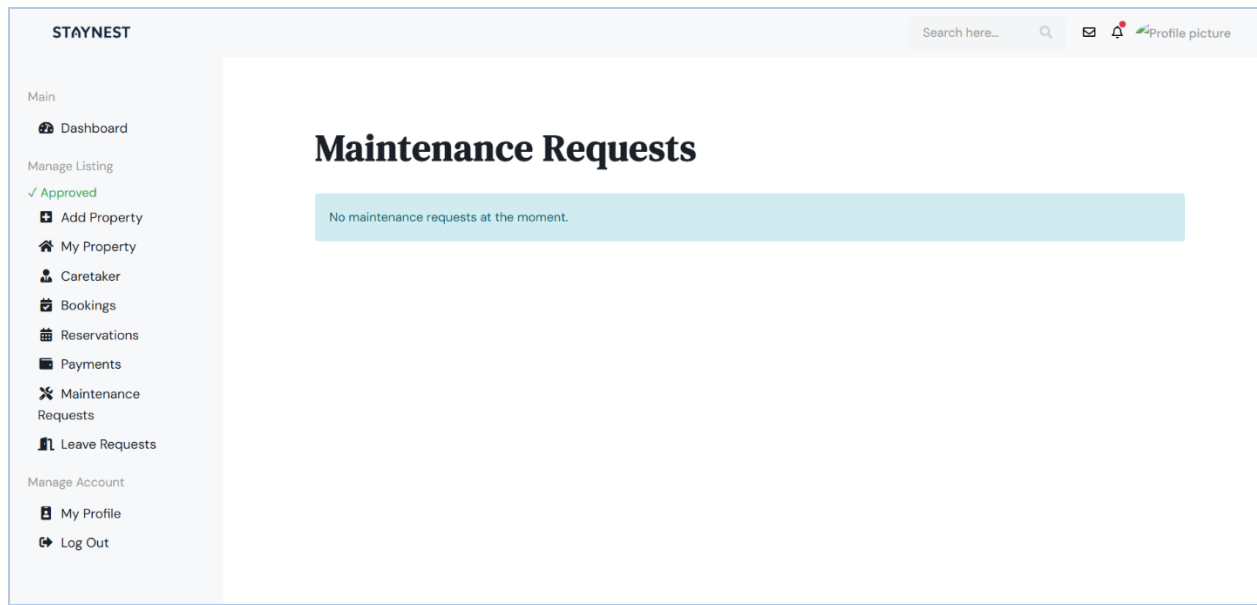


Figure 25. Maintenance Requests

Leave Requests

The leave request page tracks formal notices from tenants intending to vacate their rented units. Each request includes the tenant's name, property, reason for leaving, and expected move-out date. Owners can approve or reject requests, add remarks, and mark the request as resolved once the tenant has departed. This system ensures that departures are recorded and handled methodically, allowing owners to plan for turnover and prepare rooms for new tenants.

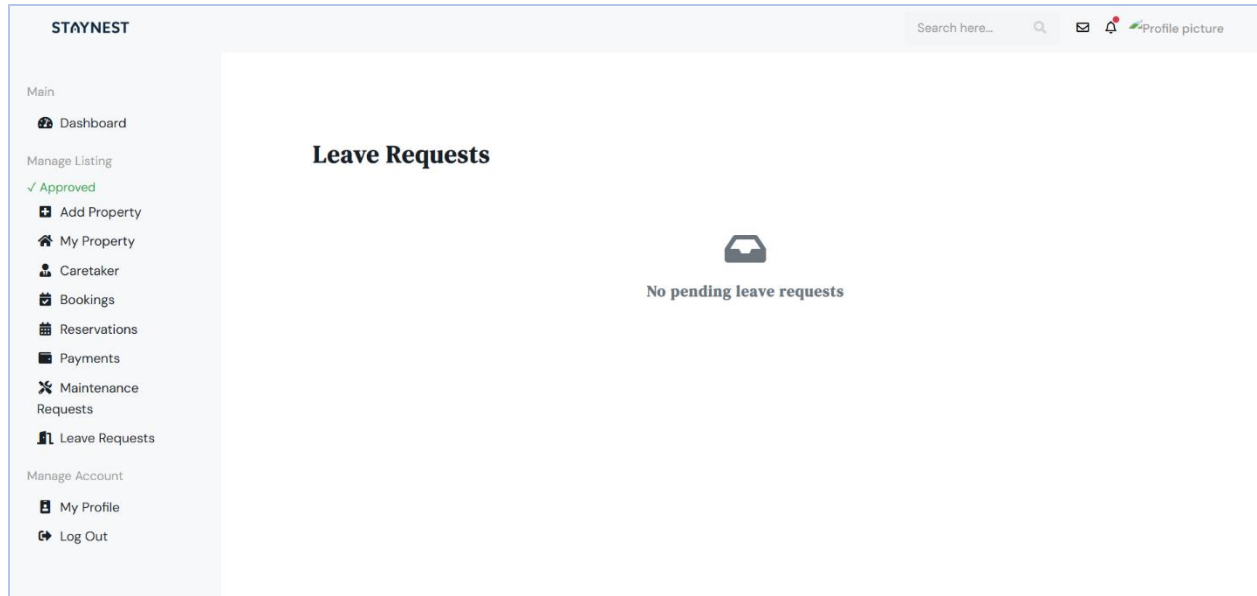


Figure 26. Leave Requests

Owner Profile

The owner profile page stores personal account information including full name, email address, phone number, and uploaded identification documents. It provides edit functionality to update contact details and change login credentials securely. The interface maintains a clean and organized layout, ensuring that owners can easily manage their personal data while upholding verification and system integrity standards.

STAYNEST

Search here...

My Profile

Basic Information

First Name * Middle Name Last Name *

Phone Number * Email *

Owner Profile

Business Name Business Address

Business Phone Business Email

Owner ID Type Owner ID Front No file chosen

Owner ID Back No file chosen

Additional info

Upload your profile photo

Note: Minimum size 300px x 300px

Figure 27. Owner Profile

Caretaker Dashboard

The caretaker dashboard displays a personalized interface tailored to caretakers assigned to one or more properties. It shows a summary of ongoing maintenance tasks, recently submitted service requests, and their status. Caretakers can view the details of each task, mark them as completed, and update progress notes. The dashboard ensures that caretakers remain informed about their responsibilities, contributing to timely property maintenance and tenant satisfaction.

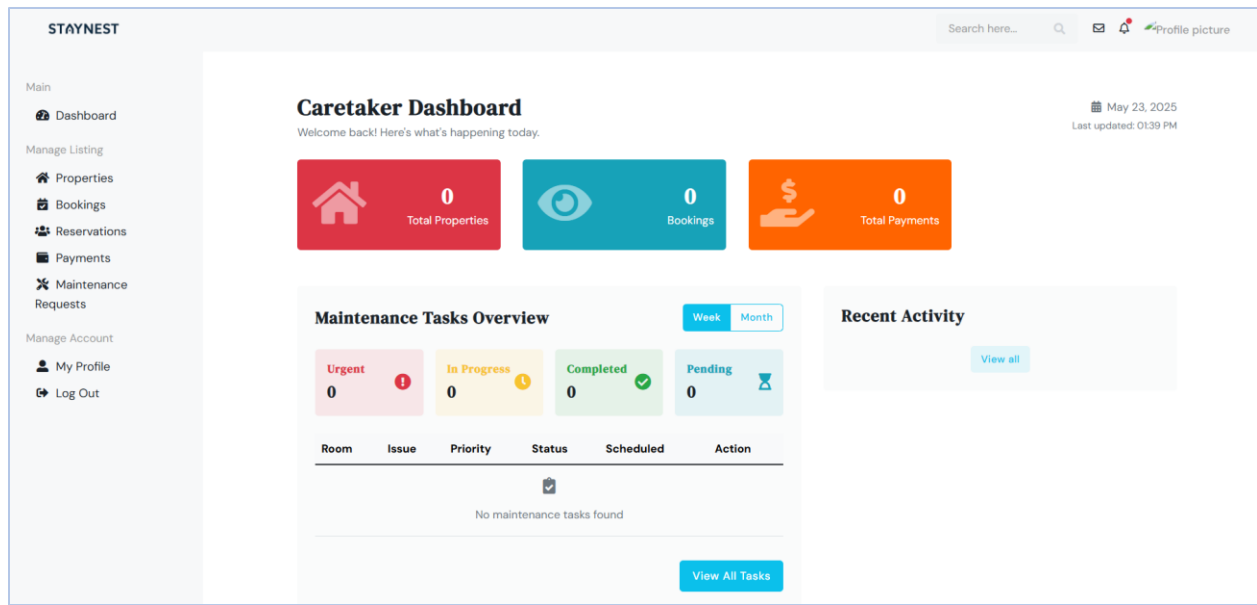


Figure 28. Caretaker Dashboard

System Architecture

Development, Testing, and Compatibility

The Development, Testing, and Compatibility section outlines the key stages involved in building the system, ensuring its reliability, and confirming that it functions properly across different platforms and devices. This includes the tools and technologies used during development, the testing methods applied to identify and resolve issues, and the compatibility checks performed to guarantee smooth performance on various hardware configurations, operating systems, and browsers. This process is essential in delivering a stable, user-friendly, and accessible application for all intended users.

Software Specifications

The Software Specifications section provides a brief overview of the system’s technical and functional requirements. It outlines the tools, platforms, and features used to ensure that the Staynest application operates efficiently, securely, and meets user needs.

Table 3. Software Requirements for Computer

Software Specifications	Minimum Requirement	Recommended Requirement
OS	Windows 10	Windows 10 or 11
phpMyAdmin	Version 4.9.11	Version 5.2.1
Bootstrap	Version 5.1	Version 5.3.0
Visual Studio Code	Version 1.66.0	Version 1.78.0
XAMPP	Version 3.3.0	Version 8.2.4
Web Browser	Chrome Version 112.0.5615121	Chrome Version 115.0.5748.0 or higher

The Software Specifications outline the system requirements needed to develop, run, and maintain the Staynest Boarding House Management and Tenant Application. At a minimum, the system requires Windows 10 as the operating system, with tools such as phpMyAdmin version 4.9.11, Bootstrap version 5.1, Visual Studio Code version 1.66.0, and XAMPP version 3.3.0 to support the backend and frontend development. For optimal performance, it is recommended to use Windows 10 or 11, phpMyAdmin version 5.2.1, Bootstrap version 5.3.0, Visual Studio Code version 1.78.0, and XAMPP version 8.2.4. The system is best accessed through a modern web browser, with Chrome version 115.0.5748.0 or higher recommended for full compatibility and smooth user experience. These specifications ensure the application operates efficiently and is aligned with current development standards.

Hardware Specifications

Hardware specifications outline the essential components and capabilities such as processor speed, memory, storage, and networking that ensure the software runs smoothly. Alongside these, software specifications define the required operating systems, development tools, and browser versions needed for optimal performance and compatibility. Together, they provide the foundation for a reliable and efficient system.

Table 4. Hardware Requirements for Smartphone

Hardware Specifications	Minimum Requirements	Recommended Requirements
Processor	Processor 2.10.12 Dual Core	Processor 2.10.12 Dual Core or higher version
Storage and Random-Access Memory (RAM)	64 GB storage, 4 GB RAM	128 GB storage, 8 GB RAM, expandable via microSD up to 1 TB
Graphics Processing Unit (GPU)	Any GPU brand with 32bit	64bit or higher version
Size and Display Resolution	2310 x 1080	2310 x 1080 or higher

The hardware specifications detail the minimum and recommended technical requirements necessary to ensure the StayNest application runs efficiently and reliably. At a minimum, the system requires a dual-core processor running at 2.10 GHz, with a quad-core CPU of 2.0 GHz for battery performance. Storage and memory requirements include at least 64 GB of storage and 4 GB of RAM, while the recommended setup doubles these to 128 GB of storage and 8 GB of RAM, with expandable storage options up to 1 TB via microSD. Graphics support requires a GPU with a minimum of 32-bit capability, with 64-bit or higher recommended for better performance. The

display should support a resolution of 2310 x 1080 pixels, with higher resolutions encouraged to enhance visual clarity. These specifications provide a balanced framework to ensure smooth operation, responsive performance, and an optimal user experience across devices.

Table 5. Hardware Requirements for Desktop Computer

Hardware Specification	Minimum Requirement	Recommended Requirement
Processor	Intel® Core™ i5 CPU 2+ GHz	Intel® Core™ i7 CPU 3+ GHz
RAM	4 GB DDR4	16 GB DDR4 or higher
Storage	256 GB SSD	512 GB NVMe SSD or higher
Graphics Card	Integrated graphics or dedicated GPU with 2 GB VRAM	Dedicated GPU with 6 GB VRAM or higher
Display	24" LCD monitor	34" LCD monitor
Power Supply Unit	400W	600W or higher with 80 Plus certification
Network Adaptor	802.11ac 2.4/5 GHz wireless adapter	802.11ac 2.4/5 GHz wireless adapter

Table 5 outlines the hardware requirements for desktop computers to ensure the smooth and efficient operation of the system. The minimum requirements include an Intel® Core™ i5 processor with a speed of 2 GHz or higher, 4 GB of DDR4 RAM, and a 256 GB solid-state drive (SSD) for basic performance. For graphical needs, either integrated graphics or a dedicated GPU with at least 2 GB of VRAM is acceptable. A 24-inch LCD monitor, a 400W power supply unit, and a standard 802.11ac 2.4/5 GHz wireless adapter complete the minimum setup. On the other hand, the recommended configuration suggests an Intel® Core™ i7 processor running at 3 GHz or above, 16 GB or more of DDR4 RAM, and a faster 512 GB NVMe SSD for improved speed

and responsiveness. A dedicated graphics card with 6 GB or more of VRAM is advised for enhanced visual performance. A larger 34-inch LCD monitor, a 600W or higher power supply with 80 Plus certification, and the same dual-band wireless adapter ensure the system runs reliably and efficiently under heavier workloads. These specifications provide a balanced foundation for both standard and advanced usage.

Programming Environment

This section outlines the programming environment used in the development of the system. It covers the essential tools, languages, and frameworks applied for both front-end design and back-end functionality, ensuring a cohesive and efficient development workflow.

Front End

This section focuses on the front-end development of the system, which involves designing and building the user interface. It highlights the technologies, frameworks, and tools used to create a responsive, interactive, and user-friendly experience for system users.

Cascading Style Sheet (CSS)



CSS is used to control the visual presentation of the system. It handles the styling aspects such as colors, fonts, spacing, and layout design. CSS enhances the overall look and feel of the application, helping maintain a consistent and user-friendly appearance across all pages.

Hypertext Markup Language (HTML)

HTML is the foundational language used to structure the content of the system's web pages. It defines the layout and organization of elements such as headings, paragraphs, forms, buttons, and links, ensuring that all content is properly arranged and accessible within the interface.

Bootstrap

Bootstrap is a front-end framework that simplifies the development of responsive and mobile-friendly interfaces. It provides a collection of pre-designed components, layout grids, and utility classes that speed up the design process and ensure that the system adapts well to various screen sizes and devices.

JavaScript

JavaScript is a scripting language used to add interactivity and dynamic features to the system. It enables functionalities such as form validation, interactive buttons, real-time updates, and smooth transitions, contributing to a more engaging and responsive user experience.

Back End

This section focuses on the back-end development of the system, which handles the server-side operations, data processing, and database management. It highlights the technologies and tools used to ensure secure data handling, efficient performance, and smooth communication between the server and the front-end interface.

PHP Language

PHP is a widely-used server-side scripting language that powers the back-end of the system. It enables dynamic content generation, handles server requests, and processes data

interactions with the database. PHP's flexibility and ease of integration with web technologies make it a popular choice for building robust and scalable web applications.

Laravel Framework

Laravel is a modern PHP framework used to streamline back-end development with its elegant syntax and powerful features. It provides built-in tools for routing, authentication, and database management, allowing developers to write clean, maintainable code while speeding up the development process. Laravel also enhances security and performance for the application.

PhpMyAdmin

phpMyAdmin is a web-based tool used for managing MySQL databases through a user-friendly interface. It allows developers and administrators to create, modify, and organize database tables, execute SQL queries, and perform backups. This tool simplifies database administration, making it easier to maintain data integrity and optimize performance.

jQuery

jQuery is a fast and lightweight JavaScript library that simplifies HTML document traversal, event handling, and animation on the client side. It enhances the interactivity of web pages by providing easy-to-use functions, enabling smoother user experiences and reducing the amount of code needed for complex tasks.

Deployment Diagram

This deployment diagram illustrates the architecture of the Staynest system. At the core of the system is the user, who interacts with Staynest through two primary interfaces: a web browser and a mobile application. The Staynest mobile app connects directly to a central server, which is responsible for processing requests and interacting with the database to retrieve or store information. Simultaneously, users can also access the Staynest.space website through a web browser, which communicates via a router and cloud infrastructure to the same backend server. This server manages business logic and ensures synchronization between the app and the website. The database, linked to the server, stores all the critical data such as user information, property listings, and booking records. The cloud and networking components, including a router, ensure reliable connectivity and scalability, allowing users to access the Staynest services from anywhere. Overall, this architecture supports a seamless user experience across platforms while maintaining centralized control and data management.

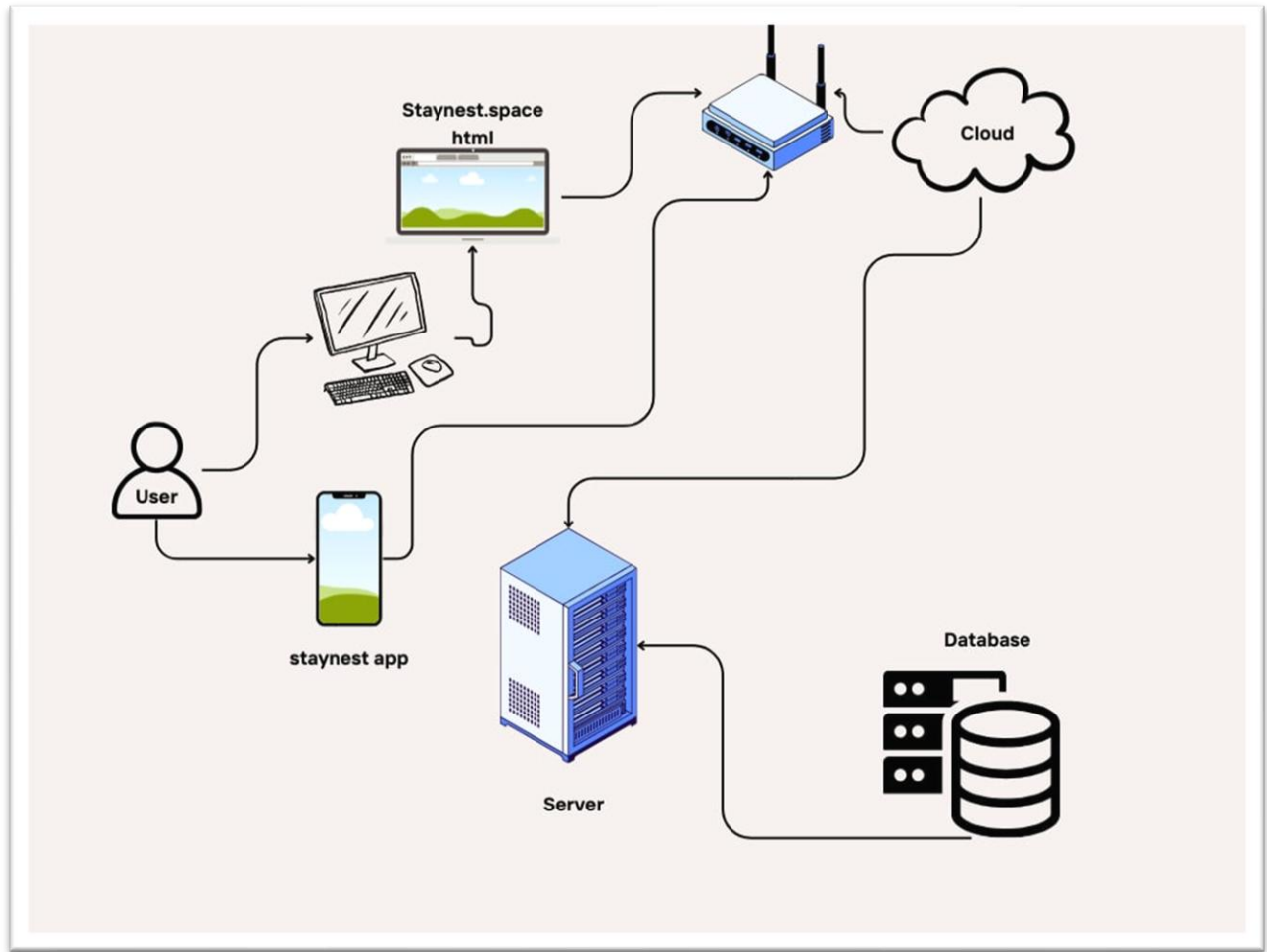


Figure 29. Deployment Diagram 

Implementation Results

This section presents the outcomes observed during the implementation of the system. It highlights how key features performed in real-world scenarios, showcases user interactions, and evaluates whether the system met its intended goals.

Unit Testing

In the unit testing phase of the Board Ease: Boarding House Management and Tenant App, each module was individually tested to ensure that its functionality and performance aligned with

the design specifications. The testing process involved the creation of detailed test cases for core features such as room reservations, tenant registration, billing and payments, and maintenance request tracking. Each test case was executed to verify that the system correctly handled various input scenarios and consistently produced the expected outputs without errors. This rigorous testing confirmed that the individual components of the system functioned accurately and efficiently, reflecting a well-structured system architecture. Any issues discovered during unit testing were promptly resolved, ensuring that each module performed reliably before progressing to the integration testing phase. This meticulous approach to unit testing played a crucial role in establishing the overall dependability and robustness of the Board Ease: Boarding House Management and Tenant App.

Integration Testing

Integration testing ensures that the individual modules and features of the StayNest Boarding House Management System work together as a cohesive whole. After each component—such as user registration, property listings, bookings, and payment processing—is developed and tested individually, they are integrated and tested as a combined unit. This process verifies that data flows smoothly between modules, interfaces function correctly when interacting, and that no conflicts or errors arise during real-world usage. Integration testing is essential to confirm that the system performs as expected when its various parts are working together, helping to identify and resolve issues early in the development process.

Compatibility Testing

Compatibility testing evaluates how well the StayNest system performs across different devices, operating systems, and web browsers. Given that users may access the platform from various mobile phones, tablets, or desktop computers, this testing phase ensures that the user interface remains consistent and functional regardless of the device or environment. The system is tested on multiple browsers such as Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge, as well as on both Android and iOS platforms. This ensures a seamless and reliable user experience for all users, minimizing display errors, functionality issues, or accessibility problems that could hinder system adoption.

Performance Testing

Performance testing measures the responsiveness, stability, and speed of the StayNest system under different conditions and workloads. It focuses on evaluating how the system behaves when multiple users access features simultaneously, such as viewing listings, submitting bookings, or processing payments. This testing identifies potential bottlenecks, ensures fast load times, and assesses the system's ability to handle high traffic without crashing or slowing down. By conducting performance testing, the development team ensures that the platform delivers a smooth, reliable experience even under peak usage, which is crucial for maintaining user satisfaction and trust.

Load Testing System

Load testing was conducted to evaluate the performance and stability of the StayNest Boarding House Management System under varying levels of user activity. The primary objective was to determine how the system responds when multiple users access and interact with its features

simultaneously, simulating real-world usage conditions. The test focused on key functions such as logging in, searching for boarding houses, booking rooms, submitting maintenance requests, and processing payments. By gradually increasing the number of simulated users, the development team was able to monitor the system's response time, throughput, and resource utilization. This testing ensured that the platform could handle high traffic without significant slowdowns or errors, confirming its readiness for deployment in an actual operational environment.

Results and Observation

The results from the load testing indicated that the StayNest system performed reliably under normal to moderately high user loads. The system maintained acceptable response times during peak interactions, with minimal delays observed in booking and payment processing. No critical failures or crashes occurred, even when user activity reached simulated peak levels. Observations also revealed that the database handled concurrent queries efficiently, and the server infrastructure remained stable throughout the testing period. Minor performance dips were noted when exceeding typical usage limits, suggesting that further optimization may be required for scalability in future updates. Overall, the system demonstrated robust performance and stability, confirming its effectiveness for real-time, multi-user operations.

Perceived Usefulness of the System According to the Tenants

Perceived Usefulness Criteria	Weighted Mean	Verbal Description
Using the System in my job would enable me to accomplish tasks more quickly.	3.77	
Using the system in my job would increase my productivity.		

Using the system would enhance my effectiveness on the job.		
Using the system would make it easier to do my job.		
I would find the system useful in my job		
Grand Mean		

This table evaluates how tenants perceive the usefulness of a given system using five specific criteria. The responses are rated on a 4-point Likert scale, with 4 representing the highest level of agreement and 1 the lowest. The table shows that the majority of tenants rated the system positively across all criteria. Specifically, questions regarding how the system enables productivity and task completion (items 1, 2, and 3) received weighted means (WM) of 3.1, while ease of job performance (item 4) slightly scored higher with a WM of 3.2. The perception of overall usefulness (item 5) also yielded a WM of 3.1. The **grand mean** across all five questions is **3.1**, indicating a generally positive perception of the system’s usefulness among tenants.

Perceived Ease-of-Use of the System According to the Tenants

Tenants rated the system’s ease-of-use across five criteria. The responses again use a 4-point Likert scale. The results were consistent with a WM of 3.0 to 3.1 across all items, resulting in a **grand mean of 3.1**. This indicates a generally favorable view among tenants regarding how easy the system is to learn, navigate, and interact with.

Perceived Ease-of-Use Criteria	Weighted Mean	Verbal Description
Learning to operate the system would be easy for me.	3.1	Useful

I would find it easy to get the system to do what I want to do.	3.1	Useful
My interaction with the system would be clear and understandable.	3.1	Useful
I would find the system to be flexible to interact with.	3.0	Useful
I would find the system, easy to use.	3.1	Useful
Grand Mean		

Perceived Usefulness of the System According to Customers

Perceived Usefulness Criteria	Weighted Mean	Verbal Description
Using the System in my job would enable me to accomplish tasks more quickly.		Very useful
Using the system in my job would increase my productivity.		
Using the system would enhance my effectiveness on the job.		
Using the system would make it easier to do my job.		
I would find the system useful in my job		
Grand Mean		

Customers provided notably high scores for system usefulness. Each of the five criteria scored a WM between 5.1 and 5.3, with the **grand mean at 5.3**. This shows a strong consensus among customers that the system significantly aids productivity, efficiency, and task accomplishment.

Perceived Ease-of-Use of the System According to Customers

Perceived Ease-of-Use Criteria	Weighted Mean	Verbal Description
Learning to operate the system would be easy for me.		Very useful
I would find it easy to get the system to do what I want to do.		
My interaction with the system would be clear and understandable.		
I would find the system to be flexible to interact with.		
I would find the system, easy to use.		
Grand Mean		

Similarly, customer responses regarding ease-of-use were highly favorable. All criteria received weighted means from 5.0 to 5.2, with the **grand mean at 5.2**. This suggests that customers found the system intuitive and flexible, reinforcing its overall user-friendliness.

Perceived Usefulness of the System According to Owners

Owners rated the system's usefulness relatively low. The weighted means across all five items ranged between 1.2 and 1.4, resulting in a **grand mean of 1.4**. This indicates a weak perception of the system's effectiveness and usefulness in their roles.

Perceived Usefulness Criteria	Weighted Mean	Verbal Description
Using the System in my job would enable me to accomplish tasks more quickly.		Very useful
Using the system in my job would increase my productivity.		
Using the system would enhance my effectiveness on the job.		
Using the system would make it easier to do my job.		

I would find the system useful in my job		
Grand Mean		

Perceived Ease-of-Use of the System According to Owners

Perceived Ease-of-Use Criteria	Weighted Mean	Verbal Description
Learning to operate the system would be easy for me.		Very useful
I would find it easy to get the system to do what I want to do.		
My interaction with the system would be clear and understandable.		
I would find the system to be flexible to interact with.		
I would find the system, easy to use.		
Grand Mean		

Ease-of-use from the owners' perspective also reflected low satisfaction. All responses yielded a WM of 1.2 to 1.4, with a grand mean of 1.4. These low scores highlight significant usability concerns from the owners' standpoint.

Perceived Usefulness of the System According to Caretakers

Perceived Usefulness Criteria	Weighted Mean	Verbal Description
Using the System in my job would enable me to accomplish tasks more quickly.		Very useful
Using the system in my job would increase my productivity.		

Using the system would enhance my effectiveness on the job.		
Using the system would make it easier to do my job.		
I would find the system useful in my job		
Grand Mean		

Caretakers provided the lowest overall ratings in both perceived usefulness and ease-of-use. For both categories, the **grand mean was 0.8**, suggesting a critical view of the system's value and user-friendliness in their roles. This indicates a strong need for system redesign or role-specific adjustments to address caretakers' needs.

Perceived Ease-of-Use of the System According to Caretakers

Perceived Ease-of-Use Criteria	Weighted Mean	Verbal Description
Learning to operate the system would be easy for me.		Very useful
I would find it easy to get the system to do what I want to do.		
My interaction with the system would be clear and understandable.		
I would find the system to be flexible to interact with.		
I would find the system, easy to use.		
Grand Mean		

This table outlines the caretakers' perceptions of the system's ease-of-use based on five specific criteria. Using a 4-point Likert scale, where 4 indicates "Very Easy to Use" and 1 denotes "Not Easy to Use," caretakers provided consistently low ratings. For all five items—ranging from

learning to operate the system to finding it easy to use overall—the weighted mean (WM) was **0.8**. This **grand mean** suggests that caretakers found the system significantly challenging to use, indicating poor usability from their perspective. These results echo previous findings and reinforce the need for usability improvements tailored to this user group.

Weighted Mean for Usability as rated by IT Experts

This table captures IT experts' assessments of the system's usability across five technical criteria: Understandability, Documentation, Buildability, Installability, and Learnability. Most criteria received high ratings, with Understandability, Documentation, and Buildability each earning a WM of 3.8, while Installability scored 3.9. Learnability, though slightly lower at 3.7, still indicates good usability. The overall grand mean **of 3.8** demonstrates a strong consensus among IT experts that the system is technically usable, particularly from a software development and implementation perspective. These findings contrast with the lower ratings from non-technical users like caretakers, highlighting a gap between technical usability and end-user accessibility.

Weighted Mean for Maintainability and Sustainability as rated by IT Experts

This figure presents IT experts' evaluations of the system's maintainability and sustainability based on thirteen technical and governance-related criteria. Categories such as Identity, Licensing, Governance, Compatibility, and Scalability were rated, with most achieving high weighted means between **3.6** and **3.9**. The highest rating (WM of **3.9**) was seen in areas like Licensing, Testability, Portability, and Reusability—indicating strong technical foundations. The lowest rating, **3.6**, was given to Identity and Governance. The **grand mean** of **3.7** suggests that

the system is considered maintainable and sustainable in the long term by IT professionals, reinforcing its readiness for future updates, integrations, and compliance.

CHAPTER V

FINDINGS, CONCLUSION AND RECOMMENDATION

This chapter provides a summary of the research conducted, presents the conclusions drawn from the findings, and offers recommendations based on the results of the study.

Summary of Findings

1. Based on our research, boarding houses in Dumaguete primarily use traditional and informal methods to reach potential customers. Many owners post their phone numbers directly on their properties so that interested clients can call to inquire about availability and rates. Others rely on social media platforms like Facebook and Messenger to provide details and communicate with people searching for a place to stay. While these methods are common, they lack a centralized system to efficiently manage bookings and inquiries
2. Designing the system was difficult because it needed to be simple and easy to use, especially since many boarding house owners were not familiar with digital systems. Most of them were used to doing things manually, like using phone calls, notebooks, or chatting online. A complicated system might confuse them or make them not want to use it. To solve this, the developers used clear and simple words, made buttons and forms easy to understand, and avoided using technical terms.
3. The Staynest Boarding House Management System, which focuses on making the process of finding and booking boarding houses faster and more convenient by showing important details such as location, amenities, prices, and the type of boarding house, was proven to be very useful. This is based on the results of our survey conducted with potential customers, where we demonstrated how the system works. Their responses

were consistently positive. The overall survey results showed that many participants rated the system as very useful and very easy to use. This proves that the developed system is both effective and beneficial for the business.

4. The Staynest Boarding House Management System was evaluated by IT experts and received high ratings for its overall usability. The experts found the system very user-friendly, noting that it is easy to understand and navigate. In terms of maintainability and sustainability, the system was also rated very positively. The IT experts confirmed that it can be effectively maintained and supported by IT staff, making it a reliable solution for long-term use.

Conclusion

1. The development of the Staynest Boarding House Management System has significantly improved the experience of both boarding house owners, potential customers and tenants. By simplifying the process of viewing, comparing, and booking available boarding houses, the system fills the gap left by traditional manual methods. The positive feedback from potential users and IT experts highlights that the system is not only easy to use but also practical and sustainable. Its user-friendly design and essential features make it a valuable tool for boarding house businesses aiming to modernize their operations and reach more clients efficiently.
2. The Staynest boarding house management system shows that even with limited resources it is possible to create a functional and helpful digital solution for local businesses. The simplified design, tailored for non-technical users, proved effective in making the system accessible while still addressing important functions needs such as

booking, information management, and customer convenience. Although some advanced features had to be excluded in the early version, the system lays a solid foundation for future improvements. With further development and support, Staynest has strong potential to become a widely adopted tool in the local boarding house industry.

3. The acceptance and positive response to the Staynest system show that even in areas where digital tools are not yet fully adopted, users are open to change when the system is simple, helpful, and easy to use. Many boarding house owners who were not familiar with technology still showed interest and appreciation for the system, especially when they saw how it could save time and make managing their business easier. This suggests that with proper support and training, communities like Dumaguete are ready to embrace digital solutions that meet their real needs.
4. Despite challenges during development such as limited resources, the need to simplify the system for non-technical users, and difficulty accessing consistent information the system was successfully built with usability in mind. Based on surveys conducted with potential users, the system was rated as very useful and easy to use. Furthermore, IT experts confirmed its usability, maintainability, and sustainability, making it a reliable solution for long-term use. Overall, the Staynest system has proven to be effective in improving how boarding house businesses operate and how customers find and book accommodations.

Recommendation

The following are the recommendations for the future enhancement of Staynest Boarding house Management System:

1. We, the researchers, recommend further improving the booking process by adding features like automated confirmations, booking history logs, and easy-to-use cancellation or rescheduling tools to make transactions smoother and more organized.
2. We, the researchers, recommend introducing real-time notifications within the system and app to inform users about booking updates, payment confirmations, and availability changes. This ensures timely communication and better user engagement.
3. We, the researchers, recommend including a messaging or chat function within the system that allows tenants to easily communicate with boarding house owners. This would enable tenants to report issues, ask questions, or request assistance quickly and efficiently.
4. We, the researchers, recommend adding a built-in feature where tenants can submit suggestions or feedback (separate from public reviews). This allows boarding house owners and system administrators to gather insights for improving service quality and system functionality.
5. We, the researchers, recommend offering training sessions and video tutorials to boarding house owners and tenants to teach them how to properly use the system's features. This will help users become more comfortable and confident with the system, encouraging better adoption and maximizing the systems benefits.

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